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UNIVERSITÀ DI BOLOGNA

Internet Data Center

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Replication is the solution for reliability and scalability

Replication → **System bottlenecks**
Single points of failure

- **Replication of server machines**
- **Replication of interconnections**
- **Replication of data**
- **Replication of application processes**
- **Replication of people**

System Engineers
Infrastructure
Computer Engineers
Managers
(one day…)

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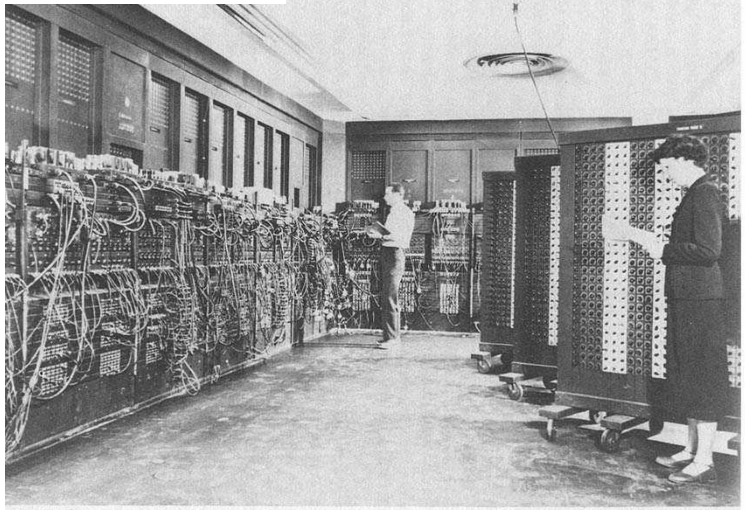
IDC is the back-end of any S&R modern service

An Internet Data Center (IDC) is the back-end infrastructure of any large company to provide <u>scalable</u> (<u>elastic</u>), <u>reliable</u>, <u>secure</u> services delivered through the Internet

The data center comes from the past with an important difference

Mainframe



The ENIAC.
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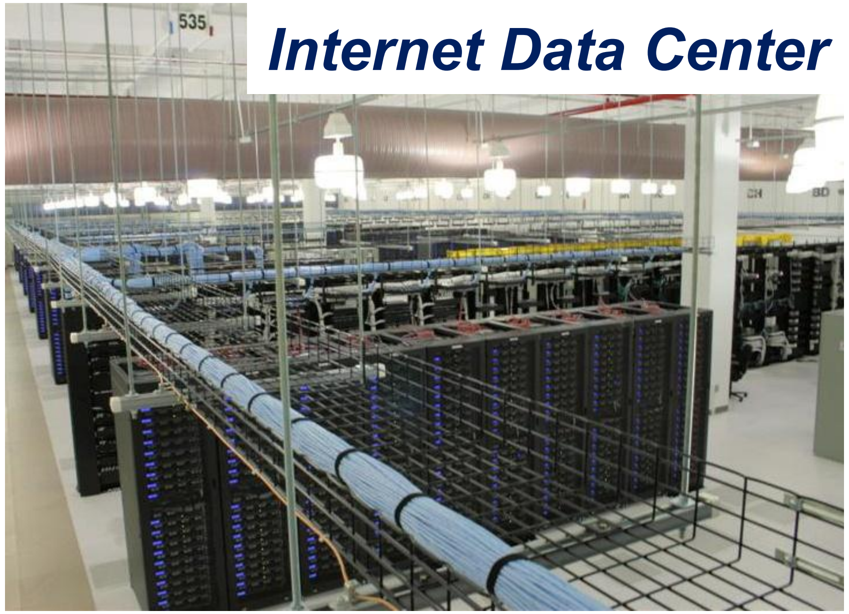
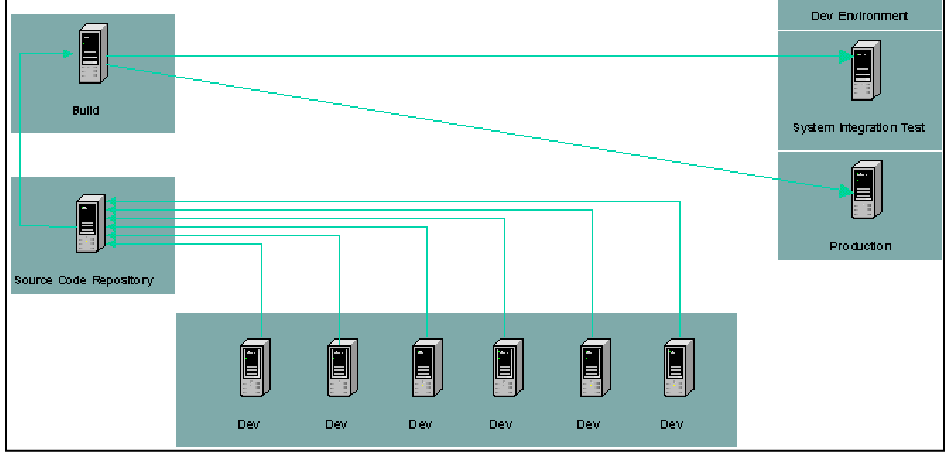
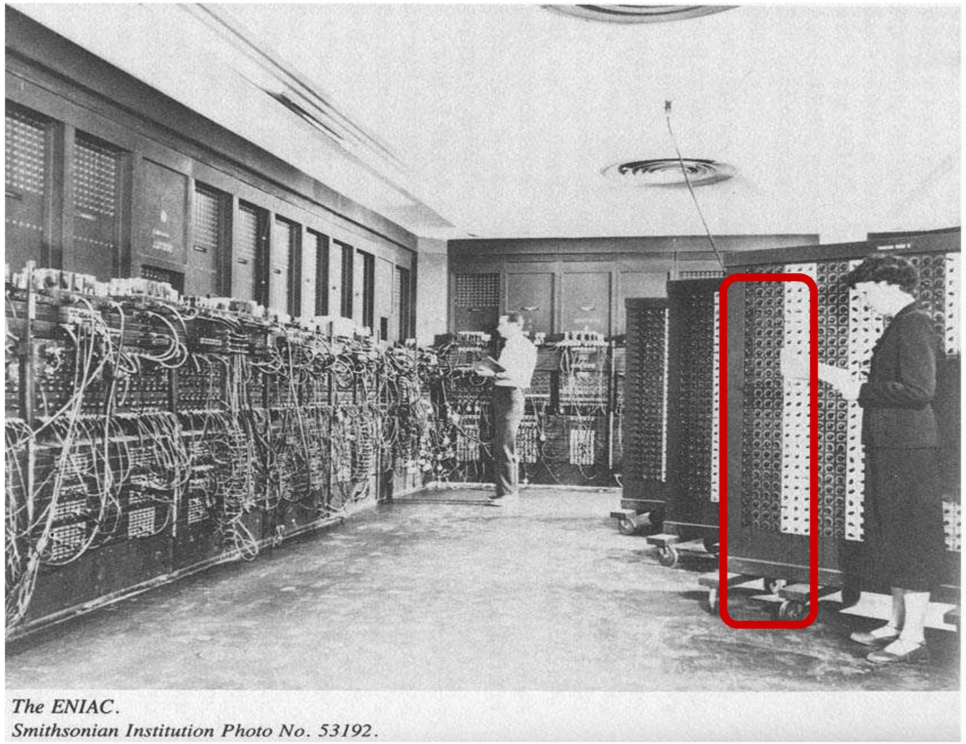
Internal Data Center



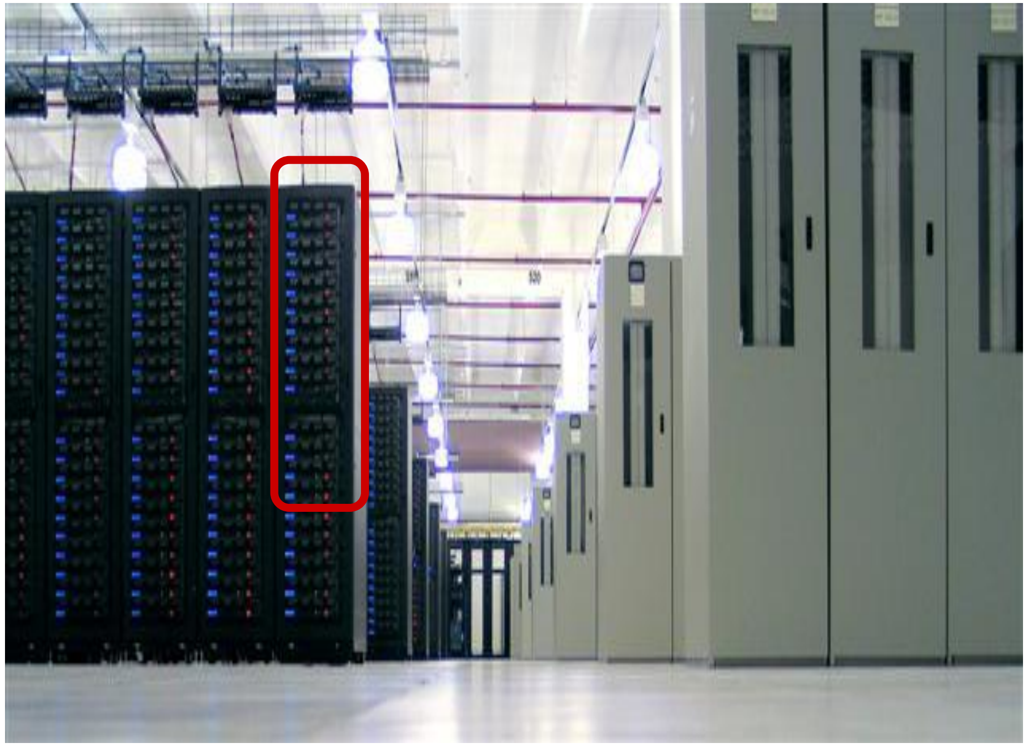
Distributed system



Similar, but with two main differences



ENIAC



Internet Data Center

Differences?

1) Internet
2) Composition of thousands of distributed server machines

The two visions

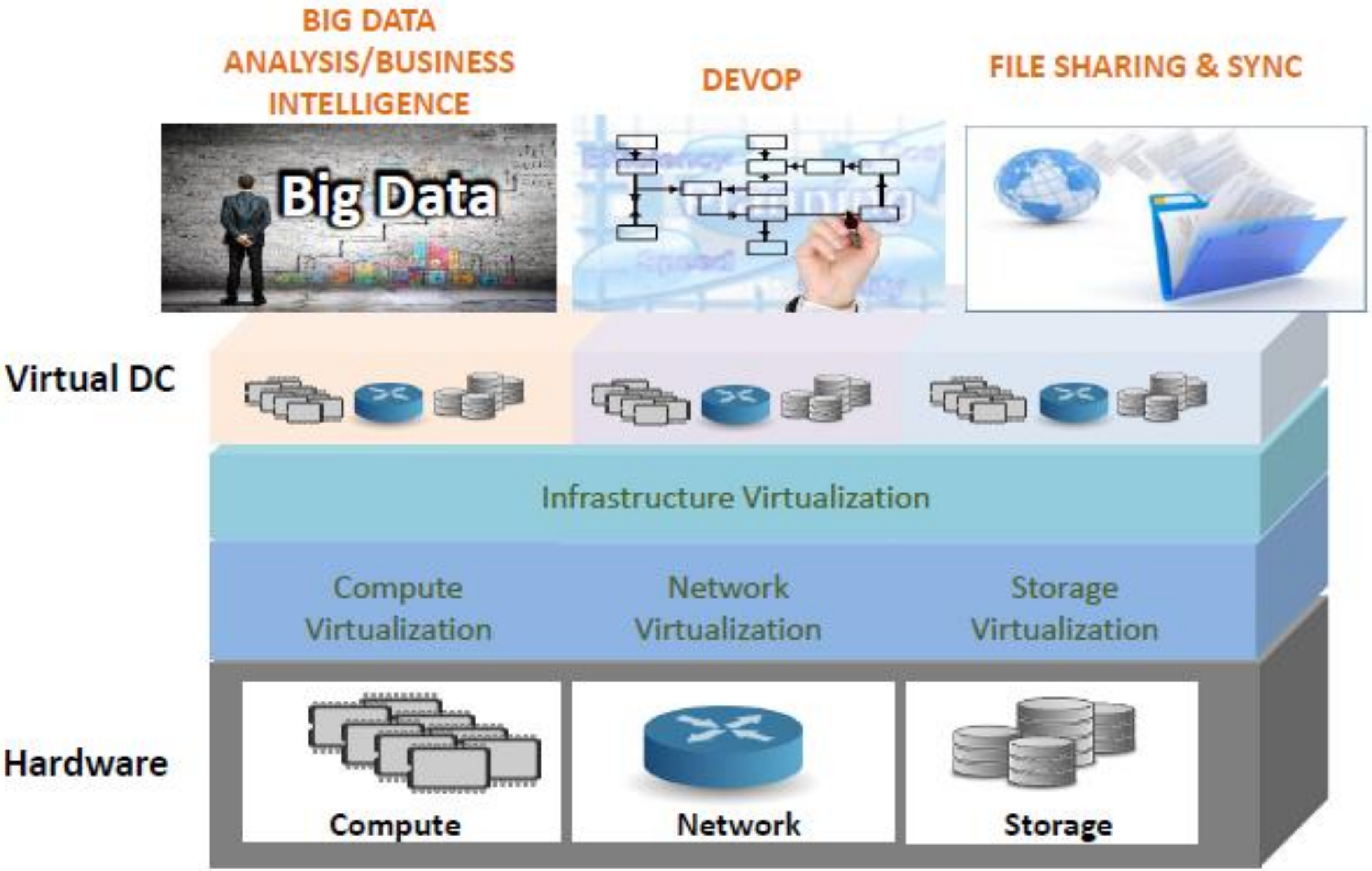
Offered vision (*User's vision*) ➔ **A logically isolated set of resources that are elastic (*scalable in two directions*), reliable, secure**

Internal vision (*Engineering vision*) ➔ **A large-scale, flexible and modular physical infrastructures consisting of**

- Racks
- PODs
- Storage systems
- Virtualization
- Advanced internal networking
- Advanced external networking

and many support infrastructures (monitoring, power, cooling, physical security, logistics, …)

The overall picture



Main components

1) Computing
2) Network
3) Data
4) Platforms
5) Applications
6) Security (logical and physical)

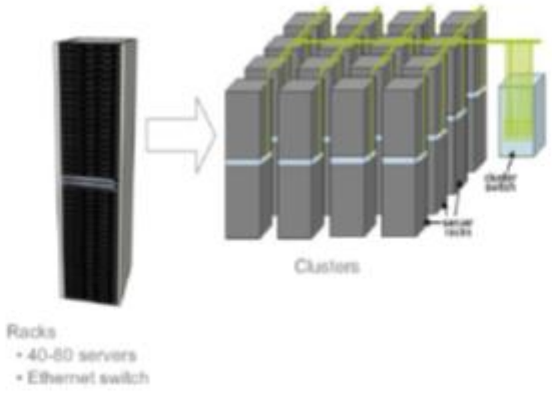
Server machine components

Basic ingredients for a Datacenter

Blade servers for computation

Rack

Interconnected racks



Basic ingredients for a Datacenter

Storage

The basics

- Disk trays
- SSD & NVM Flash

NVMe Flash

JBOD disk array

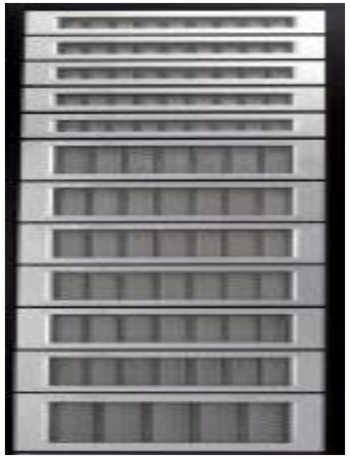
What's new

- Non-volatile memories
- New archival storage (e.g., glass)

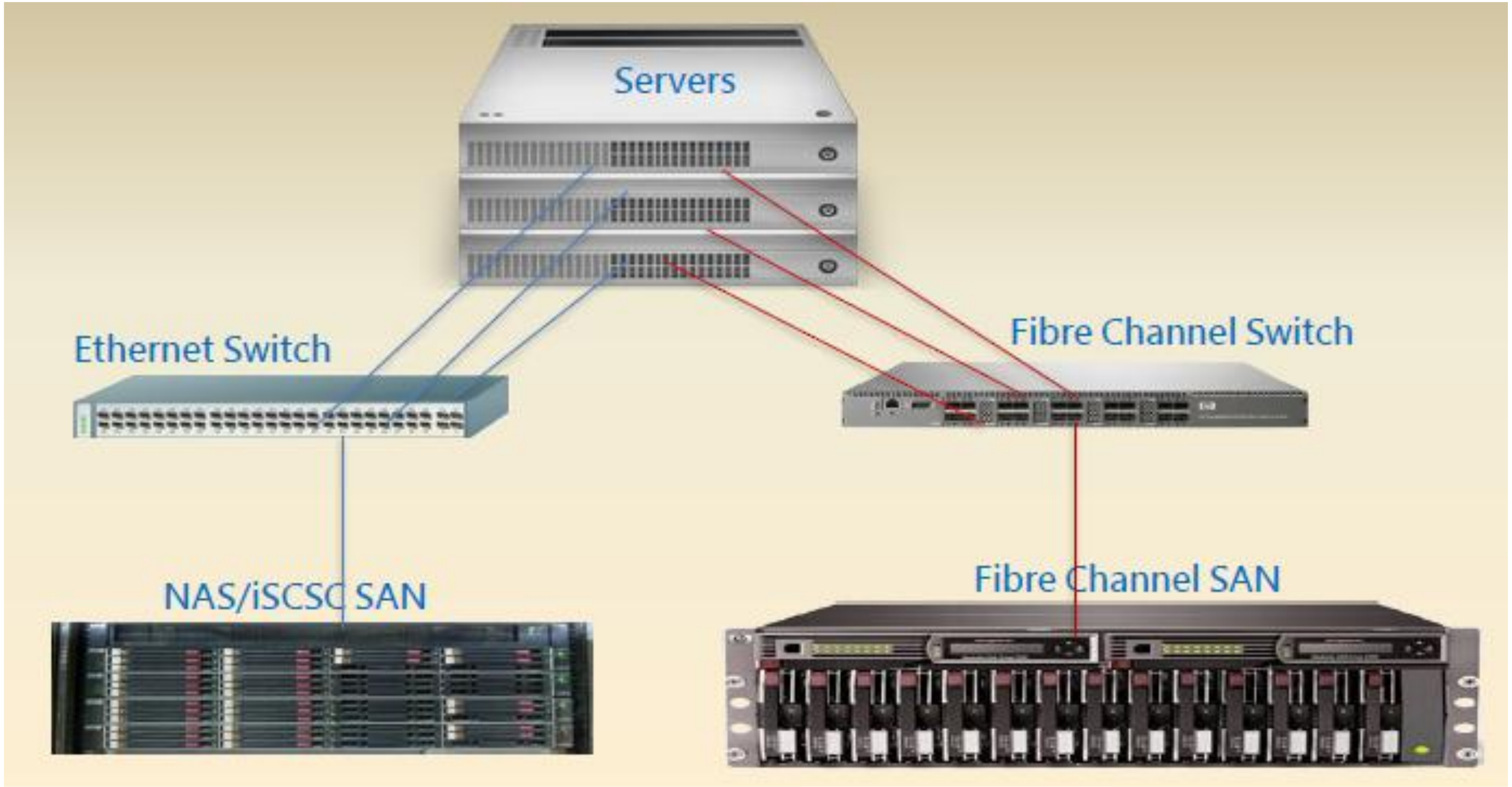
NVM DIMM

Distributed with compute or NAS systems

Blade servers for storage



Traditional connections

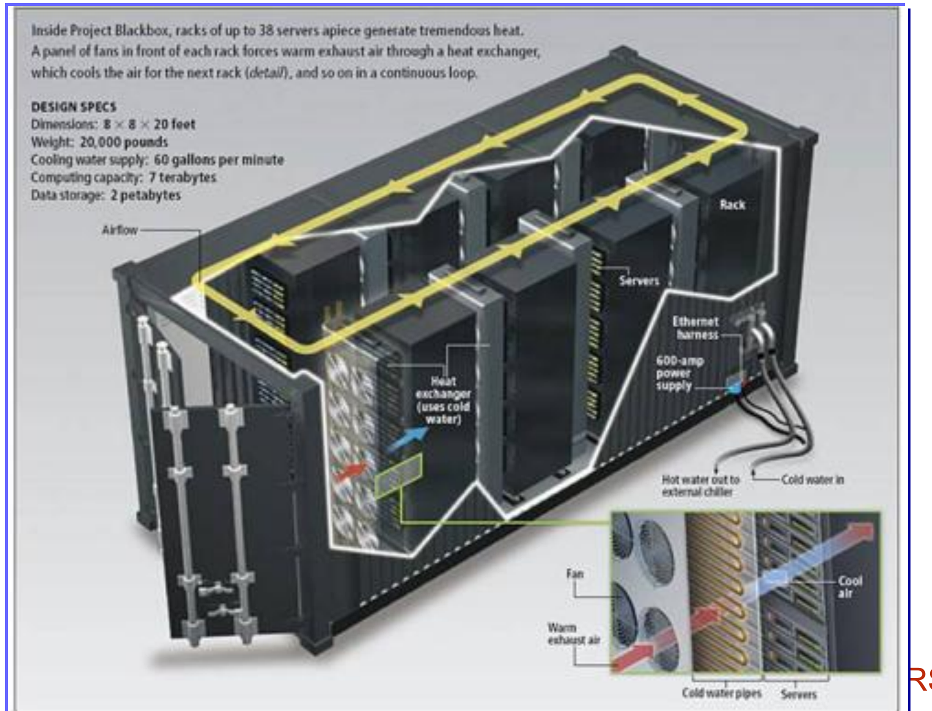


Typical datacenter

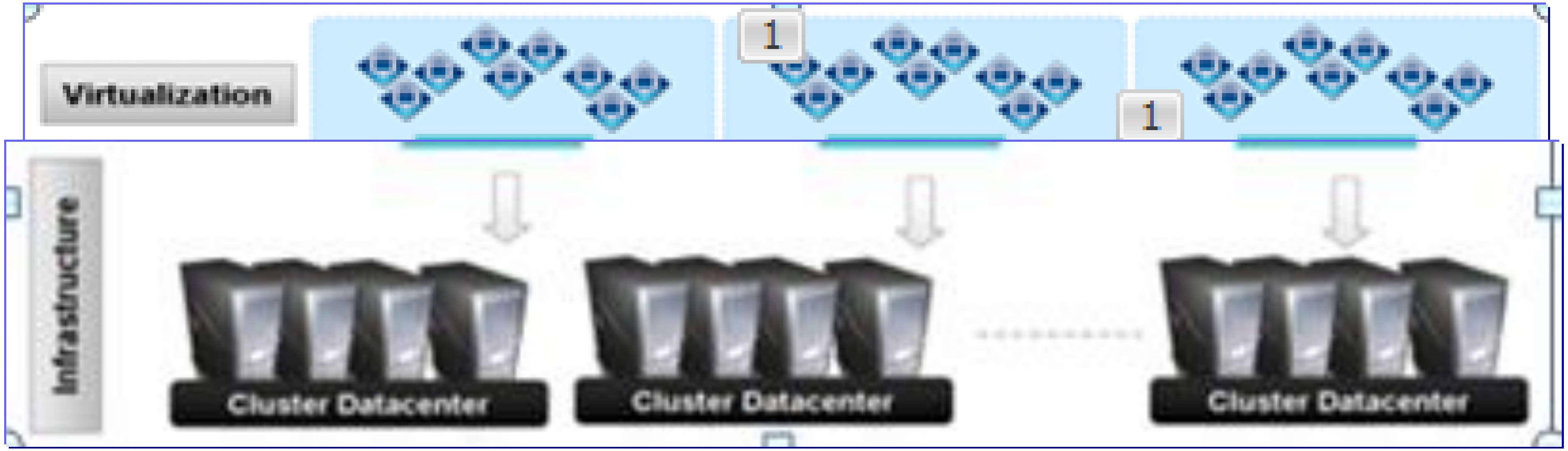
Other types: *POD*

POD = Performance Optimized Datacenter

- A container-type data center, including multiple racks (10-20), data storage devices, network infrastructure, power plant, cooling system
- Modular approach to data center expansion that uses preconfigured components to provide extra power capacity where it is needed



Virtualization is the fundamental ingredient of all modern data centers

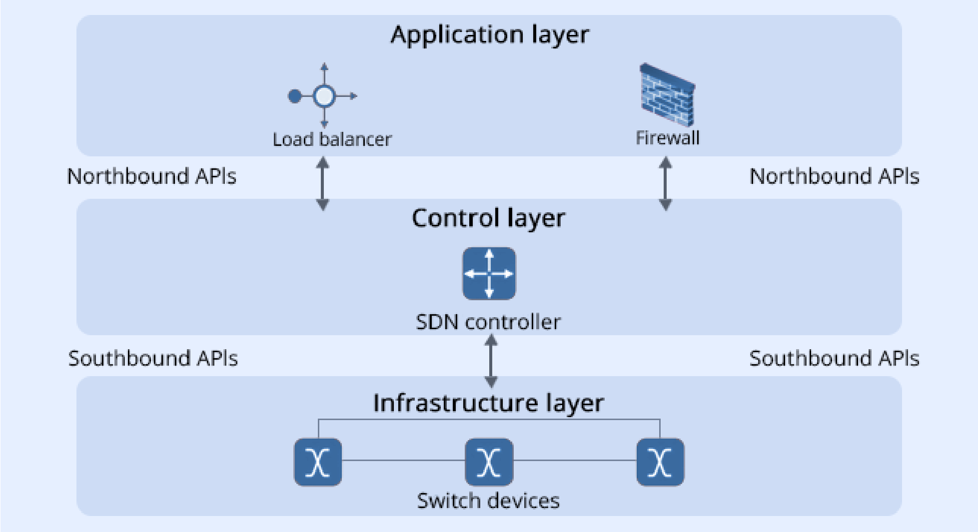


Internal network

Main network components

- Switches
- Routers
- Load balancers
- Firewalls

The shift to virtual → ***Software Defined Network*** (SDN) that moves switch management from the physical port to software that follows the workload of the virtual machines

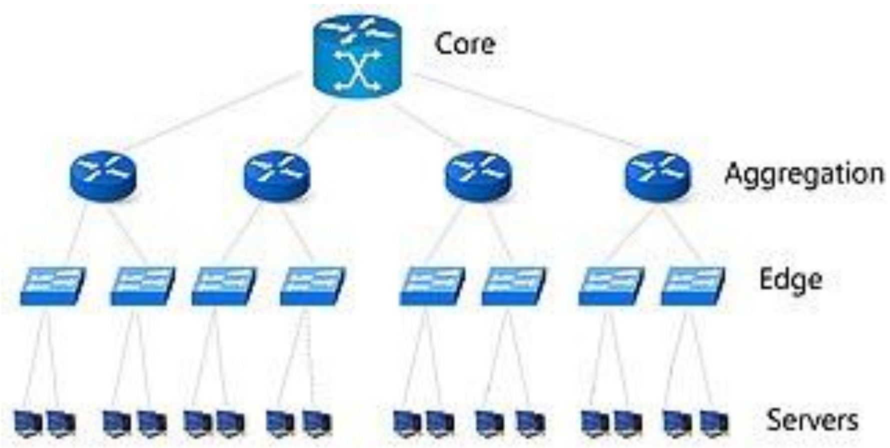


The advantages of the *internal* network

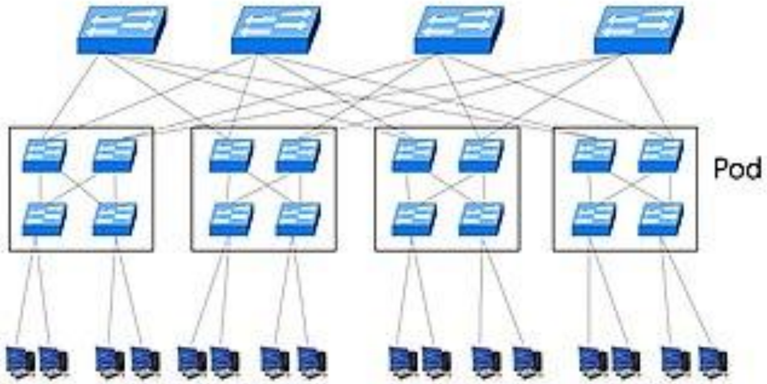
- **The internal networks of data centers are not mini-Internets that are under your control. They can be designed and managed (*if you know your requirements*)**

- **Facilitators:**
  - **single owner/administration domain** [*the most important difference with GRID computing*]
  - **limited hardware/software diversity**
  - **no malicious or selfish node**
  - **cloud providers know (and can define) the best topology in terms of performance, availability, scalability**

Data Center networking (1st generation)



Basic Tree



Fat Tree

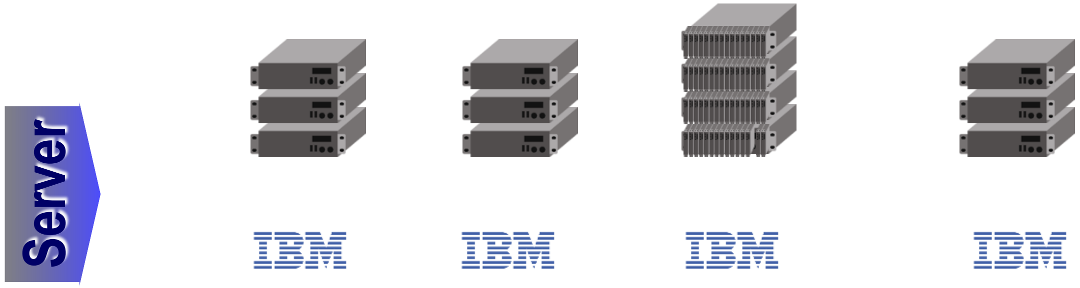
Data Center networking (1^{st} generation)

1 GbE

4 Gbps FC

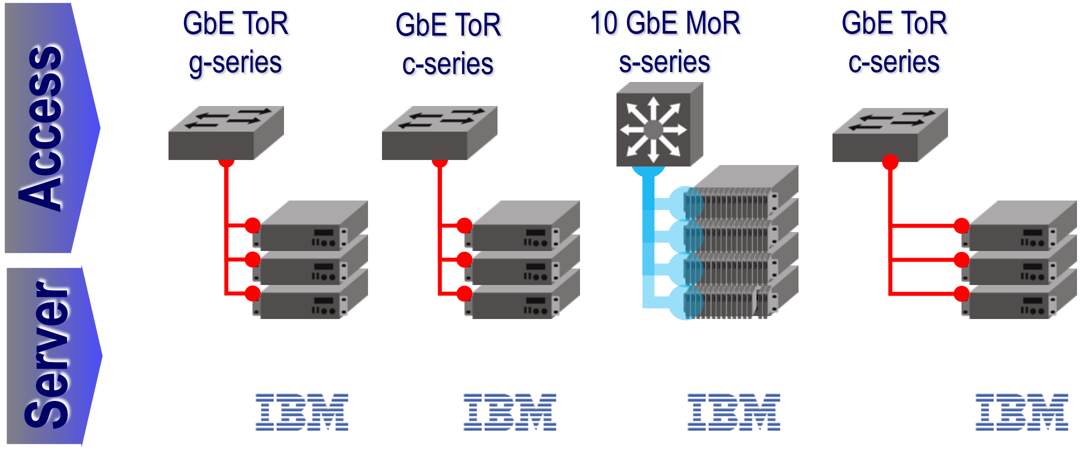
8 Gbps FC

10 GbE



Data Center networking (1st generation)

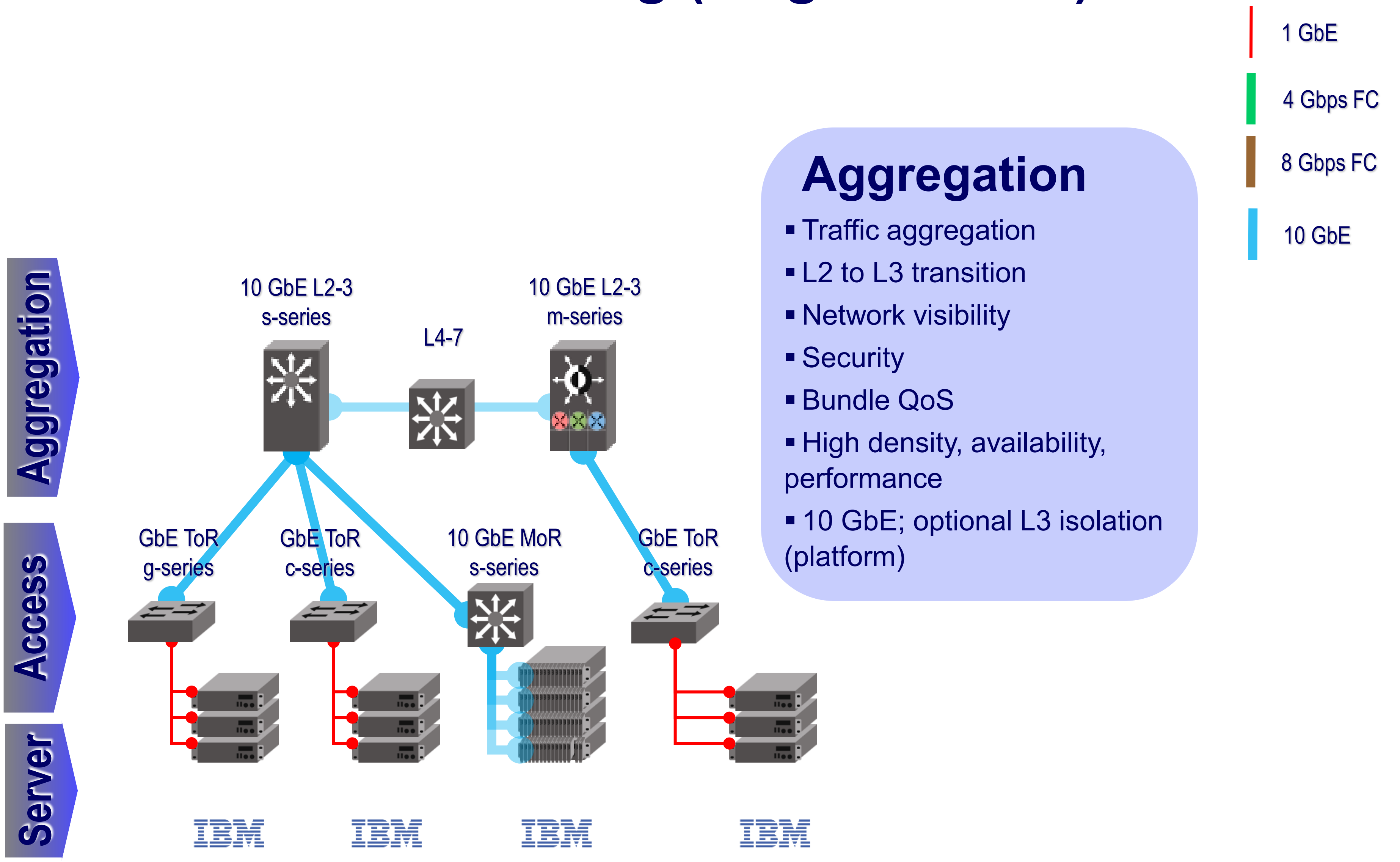
- 1 GbE
- 4 Gbps FC
- 8 Gbps FC
- 10 GbE



Access

- Server connectivity
- Traffic isolation (L2 and L3)
- Security and QoS policies
- Traffic and fault management
- GigE, 10GigE; fixed-port and modular

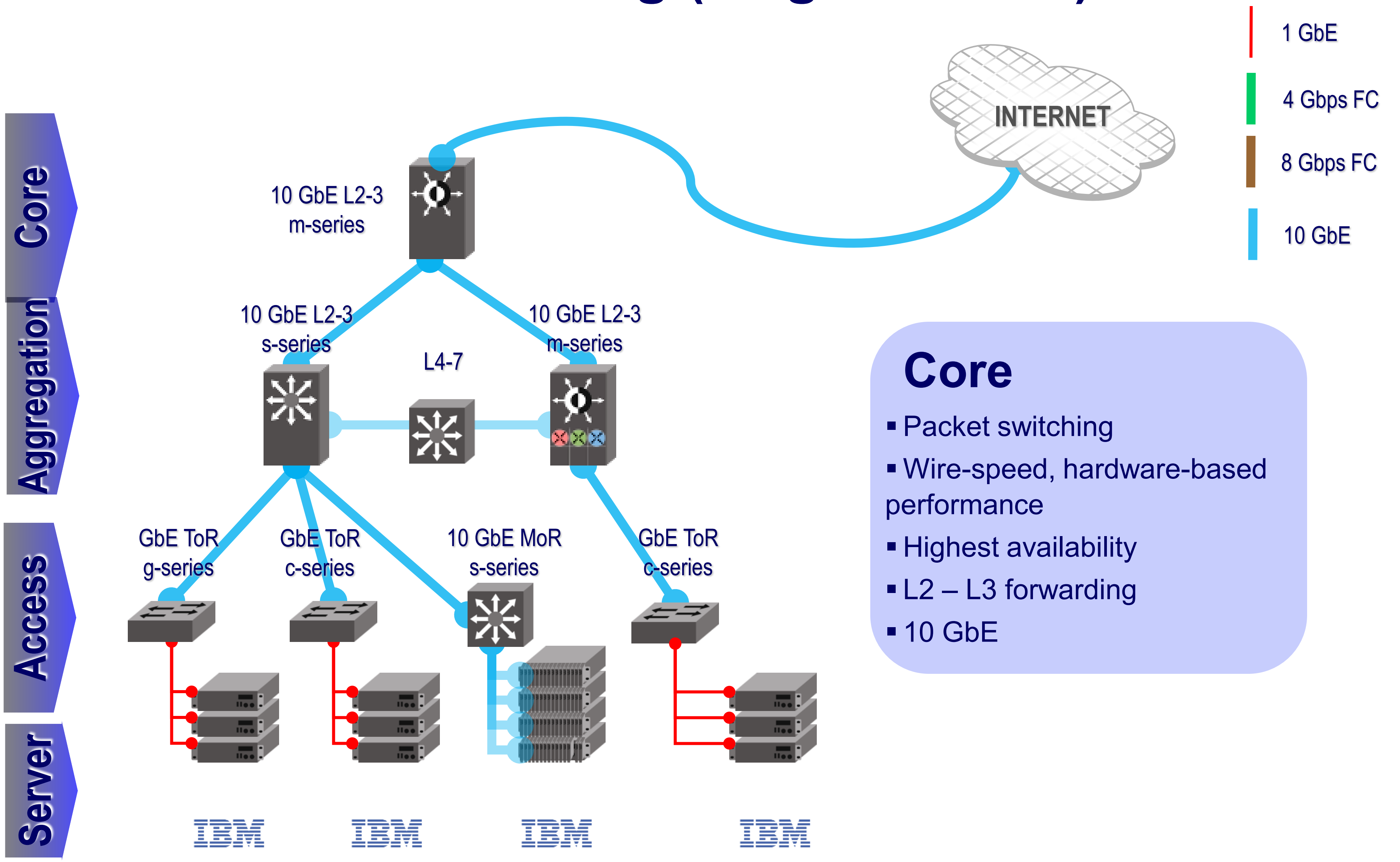
Data Center networking (1st generation)



Aggregation

- Traffic aggregation
- L2 to L3 transition
- Network visibility
- Security
- Bundle QoS
- High density, availability, performance
- 10 GbE; optional L3 isolation (platform)

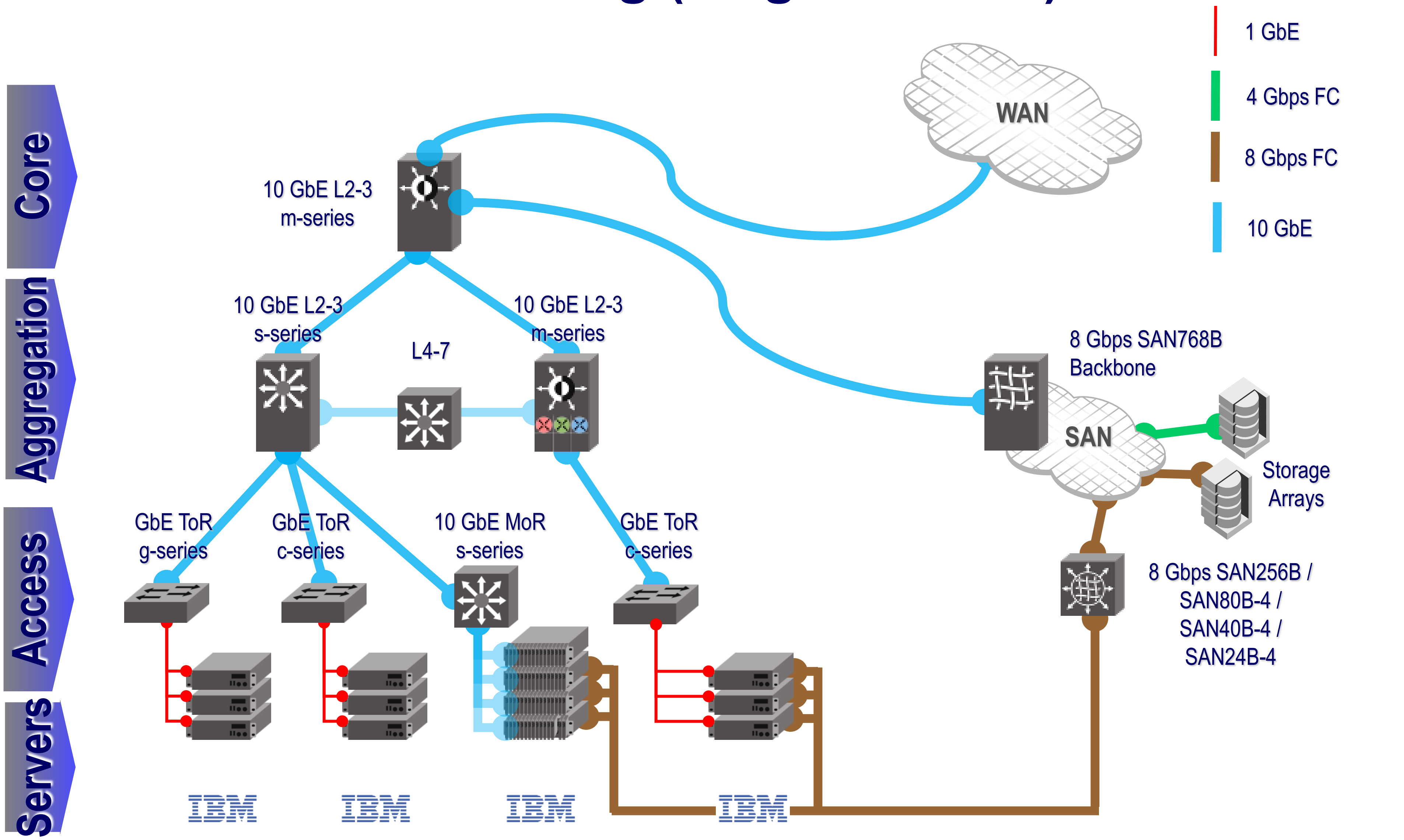
Data Center networking (1st generation)



Core

- Packet switching
- Wire-speed, hardware-based performance
- Highest availability
- L2 – L3 forwarding
- 10 GbE

Data Center networking (1st generation)



MAIN GOAL for modern datacenter → *Agility*

- **Agility inside one datacenter → any virtual server can be dynamically assigned to any service anywhere in the data center**
- Unfortunately, conventional datacenter network designs work against agility by ***fragmenting both network and server capacity***, and limiting ***elasticity*** (=the dynamic growing and shrinking of the server pools)
- Multiple applications run inside a single datacenter, typically with each application hosted on its own set of (potentially virtual) server machines

Datacenter traffic

A datacenter network supports two types of traffic:

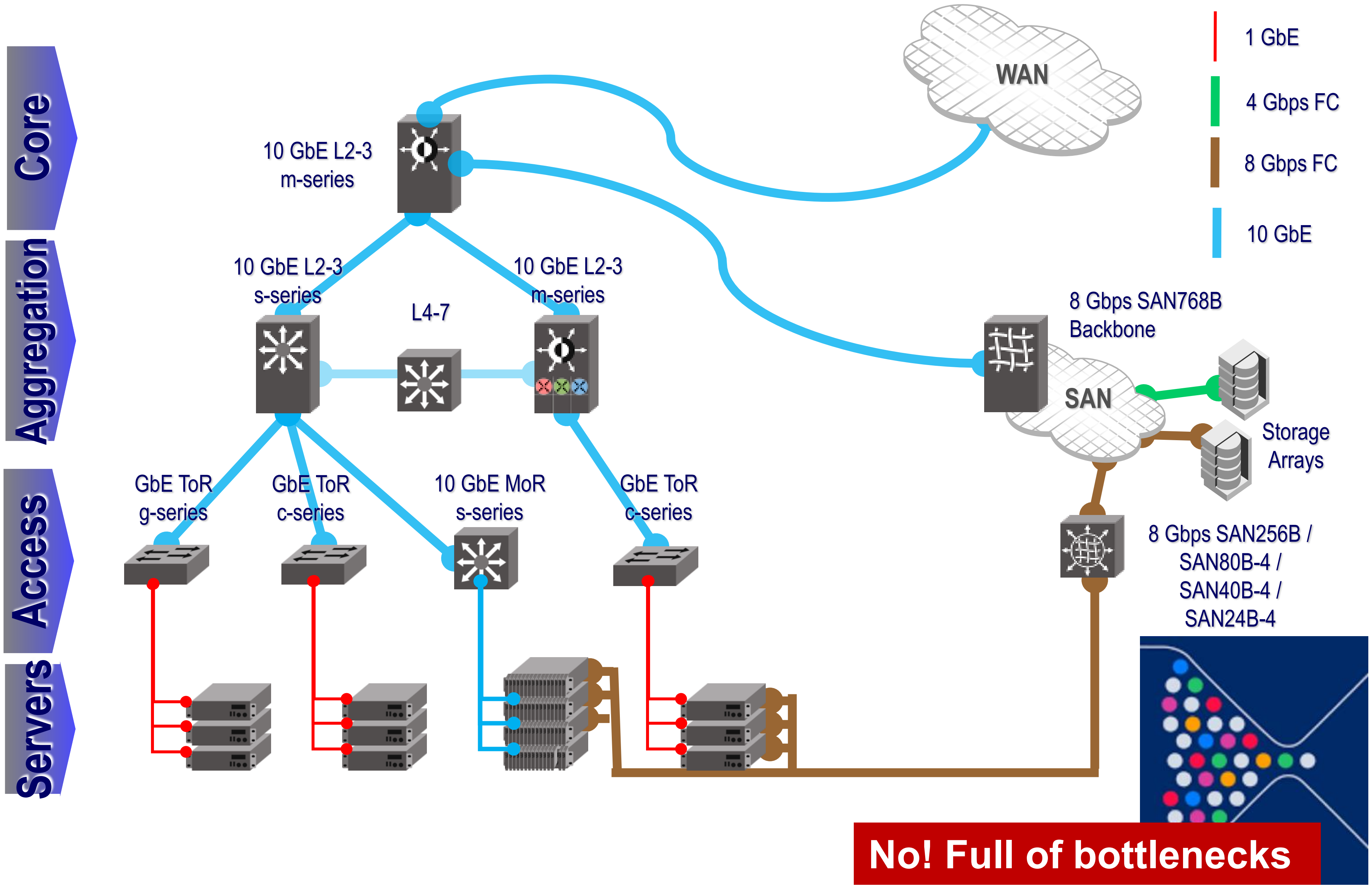
a) **Traffic flowing between external systems and internal servers**. For example, in *video download* applications, external traffic dominates

b) **Traffic flowing among internal servers**. For example, in *search applications* and *customized responses*, internal traffic dominates because of building and synchronizing instances of the indexes, and inter-process communications

Often, you have to manage both requirements:

1) low Internet latency → get access to hundreds of ISPs
2) any-to-any internal interconnection

Can 1st generation topologies work today?



No! Full of bottlenecks

In-Out traffic

- To support external requests from the Internet, an application is associated with one or more publicly visible and routable IP address(es) to which clients send their requests and from which they receive replies

- Inside the Data Center, the requests are spread by a ***specialized hardware load balancer*** among a pool of front-end servers that process the requests

- The IP address to which requests are sent is called a **virtual IP address** (**VIP**)
- The IP addresses of the servers over which the requests are spread are known as **direct IP addresses** (**DIP**)

Solutions for agility

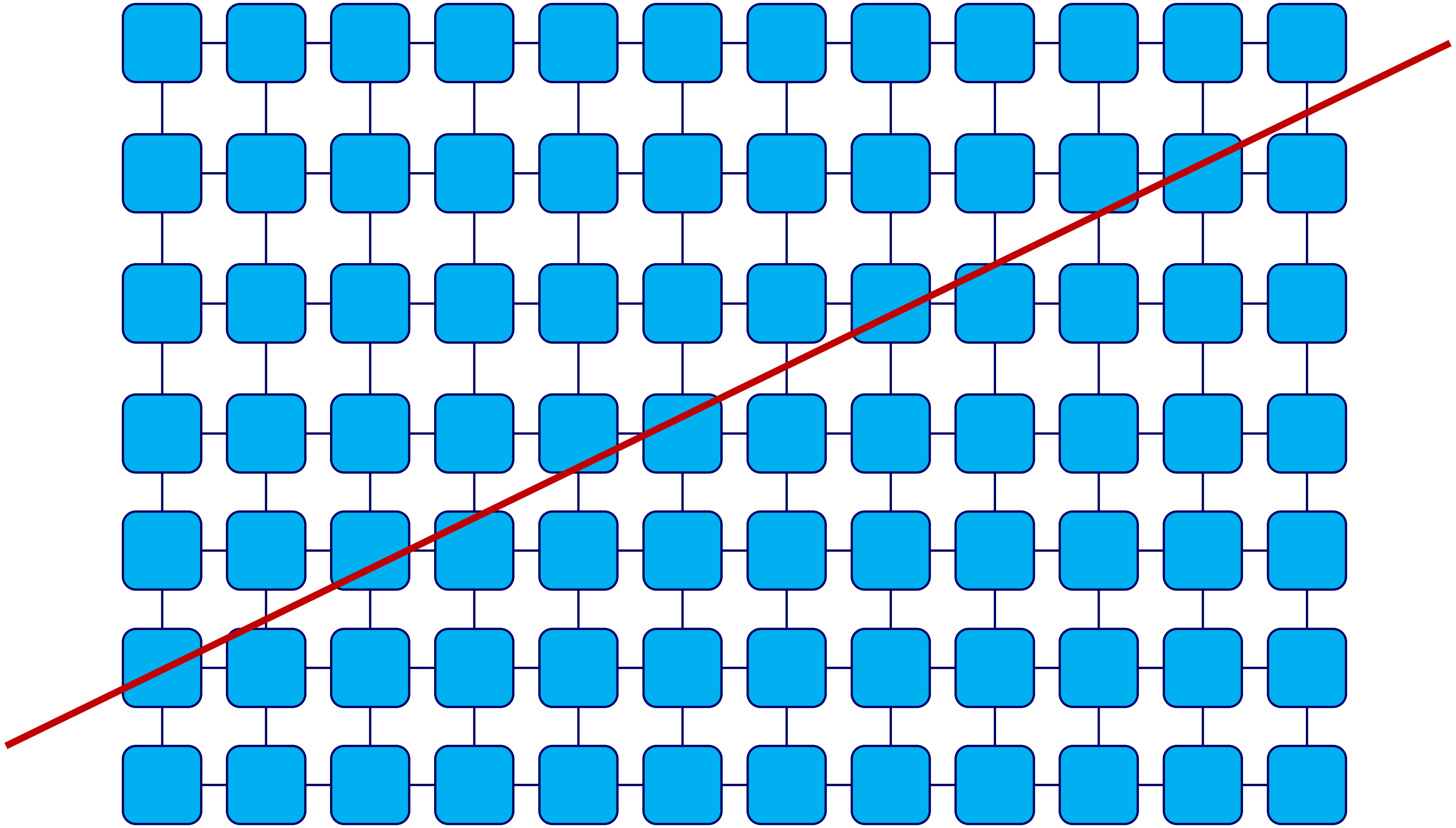
Location-independent addressing

- Services should use location-independent addresses that decouple the server's location in the datacenter from its address. This could enable any server to become part of any server pool while simplifying configuration management

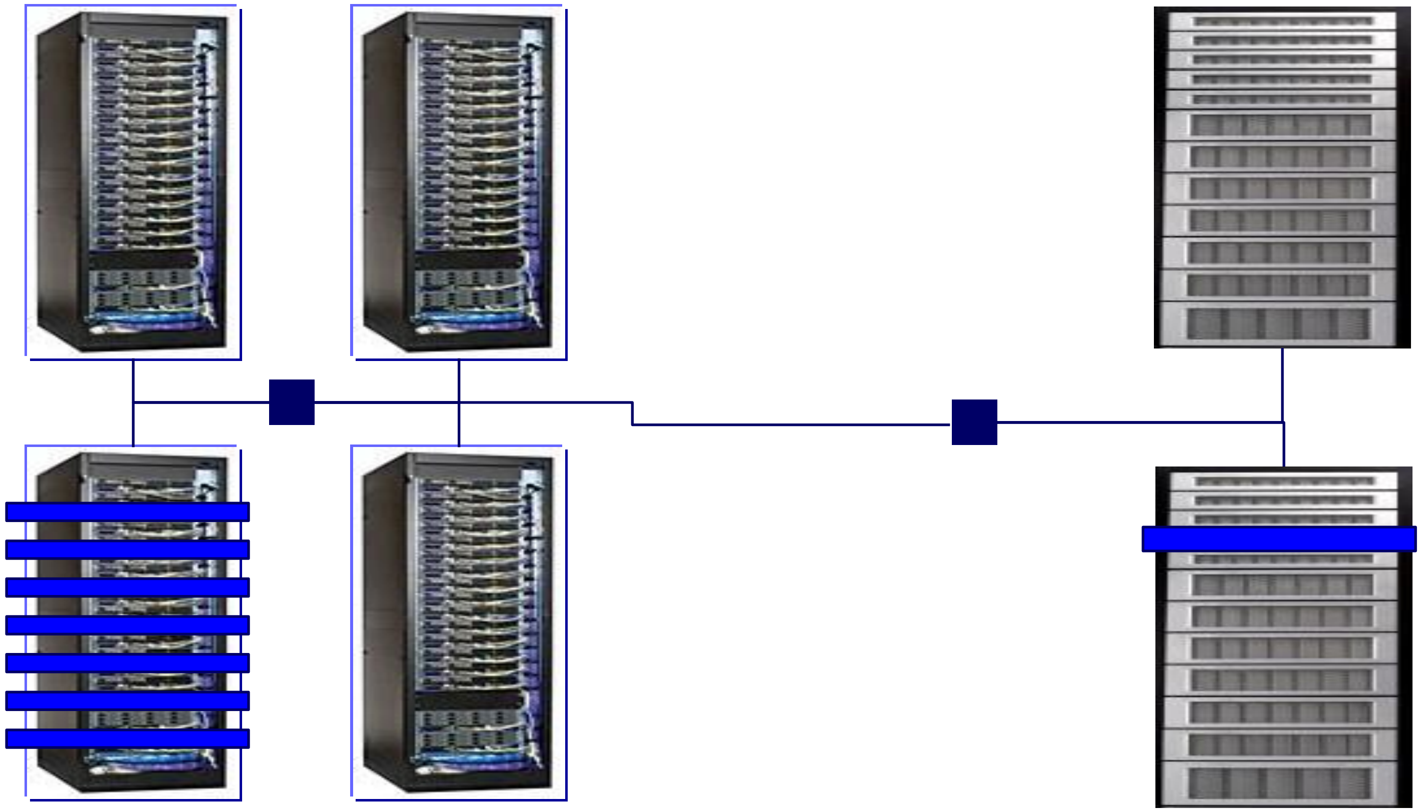
Uniform bandwidth and latency

- If the available bandwidth between two servers is not dependent on where they are located, then the servers for a given service could be distributed arbitrarily in the data center without fear of running into bandwidth bottleneck points
- Uniform bandwidth, combined with uniform latency between any two servers would allow services to achieve same performance regardless of the location of their servers

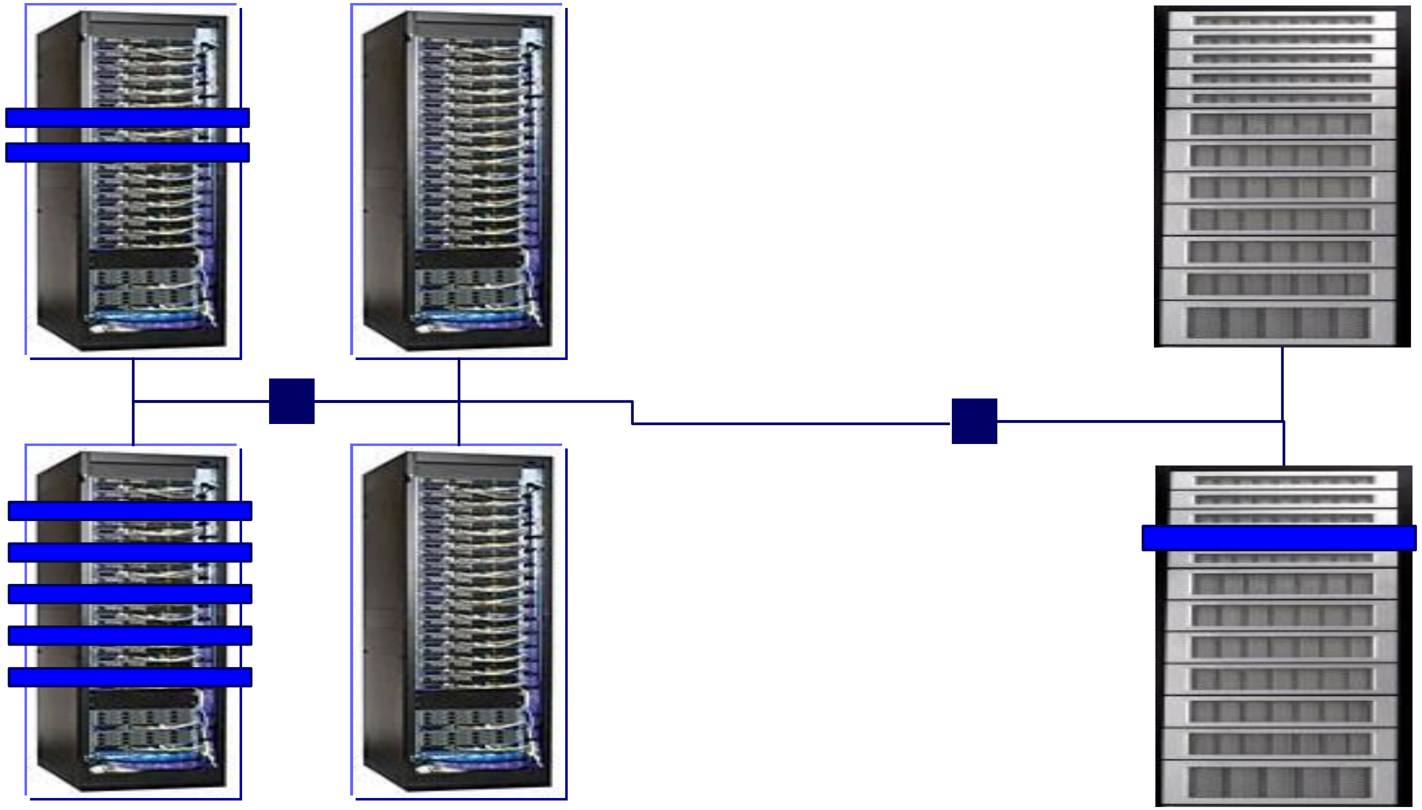
Server resources in a datacenter based on switches and routers do not represent a uniform pool



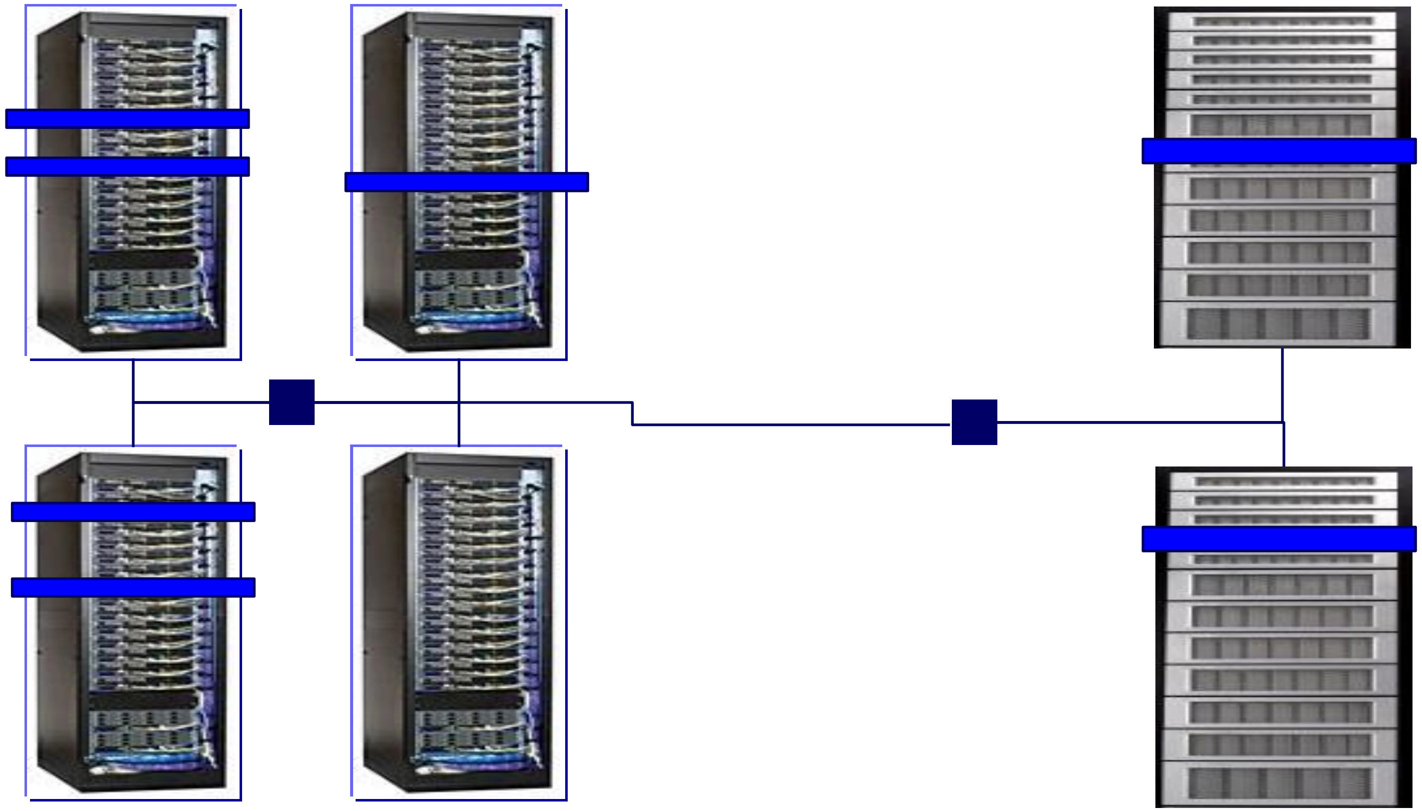
Everything is virtual and easily reconfigurable.
Example, Your application at time T0



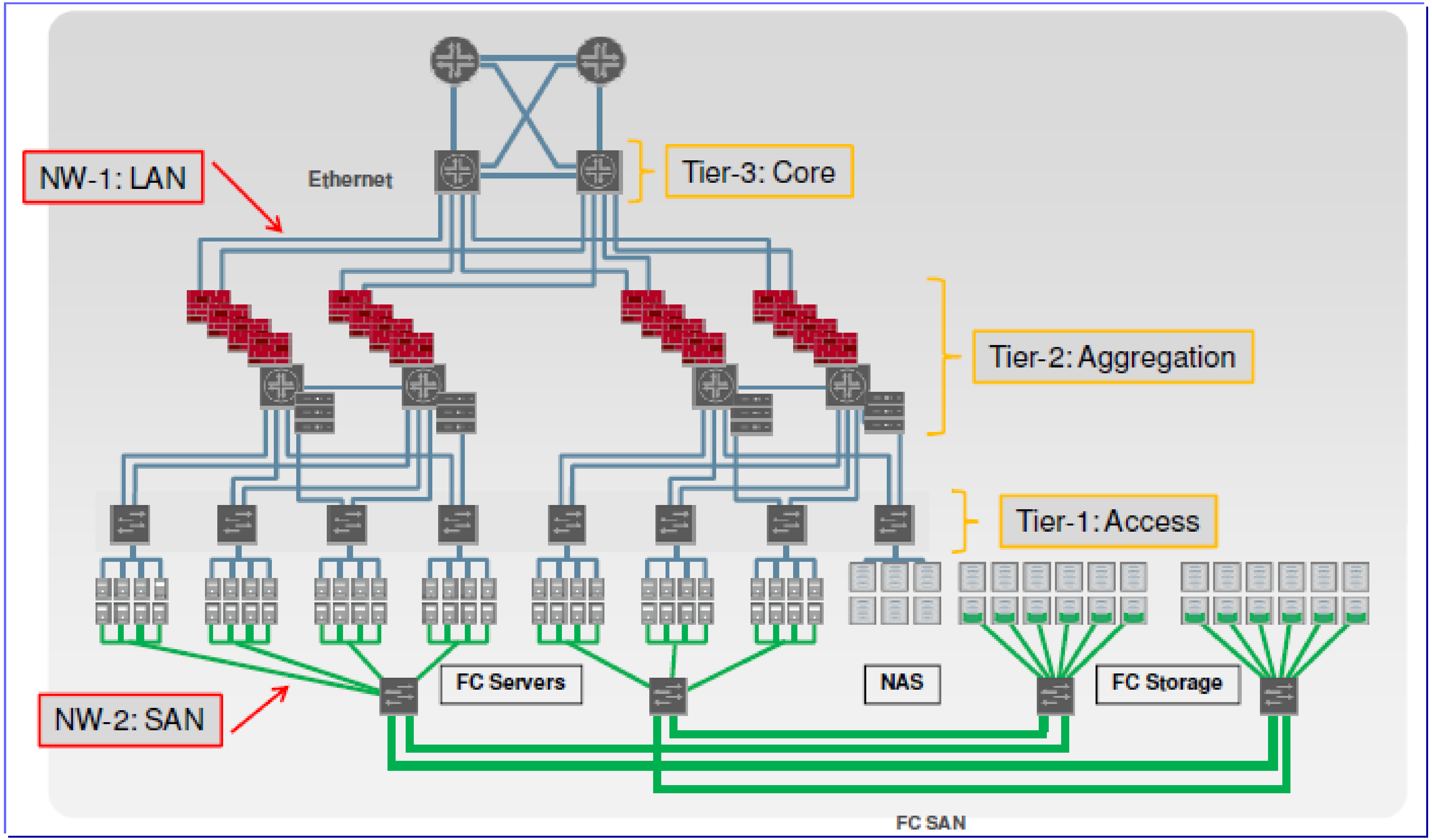
Your application at time T1



Your application at time T2



Duplication in internal networks is not enough



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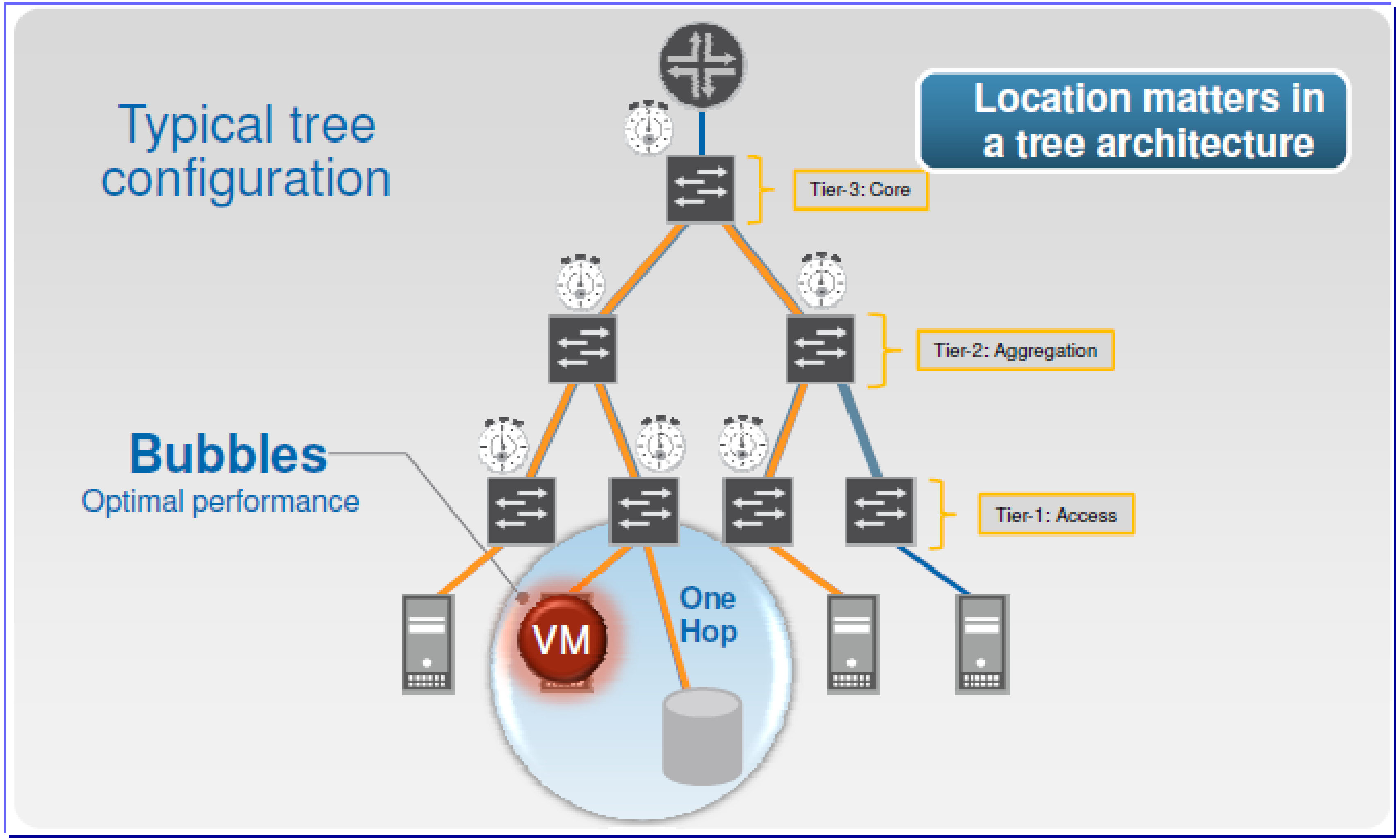
Location matters (and this is not good for performance)



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Problem 1: *Static network assignment*

- To support internal traffic within the datacenter, individual applications are mapped to specific physical switches and routers, relying heavily on VLANs and layer-3 based VLAN spanning to cover the servers dedicated to the application
- Positive consequences:
  - High degree of performance and security isolation
- Negative consequences:
  - VLANs are often **policy-overloaded**, integrating traffic management, security, and performance isolation
  - VLAN and large server pools **concentrate traffic on links high in the tree**, where links and routers are highly overbooked

Problem 2: Fragmentation of resources

Some load balancing techniques (half NAT) require that all IPs in a pool of Virtual IP be in the same layer 2 domain

- PROBLEM: If an application grows and requires more front-end servers, it cannot use available servers in other layer 2 domains - ultimately resulting in fragmentation and under-utilization of resources

Load balancing via full NAT allows servers to be spread across multiple layer 2 domains

- PROBLEM: the servers do not see the client IP, which is unacceptable because servers use the client IP for everything, from data mining and response customization to regulatory compliance

Problem 3: Poor server-to-server connectivity

The communication between servers in different layer 2 domains must go through the layer 3 network portions

- Layer 3 ports are significantly more expensive than layer 2 ports (cost of supporting large buffers, marketplace factors). As a result, these links are typically oversubscribed by factors of 10:1 to 80:1 ➔ *The bandwidth available between servers in different parts of the datacenter can be limited*
- *Managing the scarce internal bandwidth is an interesting global optimization problem*: servers from all applications must be placed with great care to ensure that the sum of their traffic does not saturate any of the network links → Unfortunately, achieving this level of coordination between continuously changing applications is impossible in practice

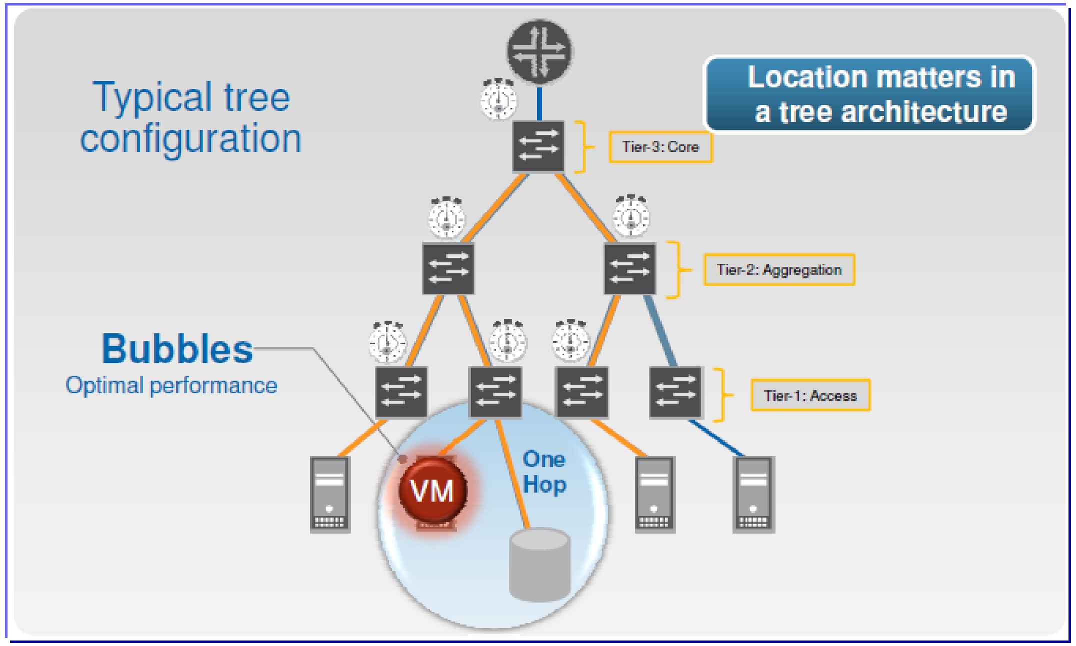
Technical solutions for agility

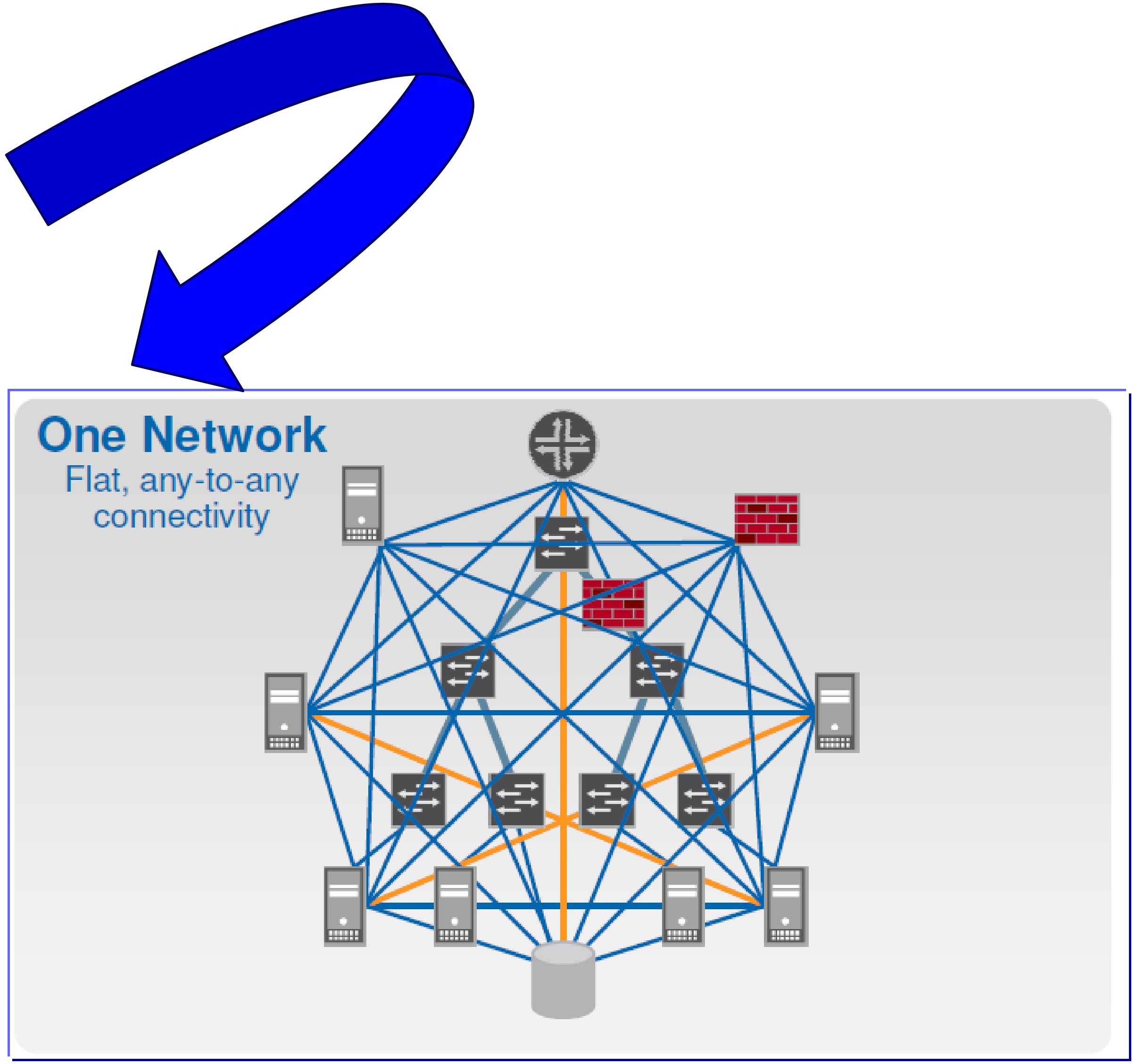
Several approaches are being explored to meet the requirements of *agility* in intra datacenter networks

- **Location-independent addressing**
- **Uniform bandwidth and latency**

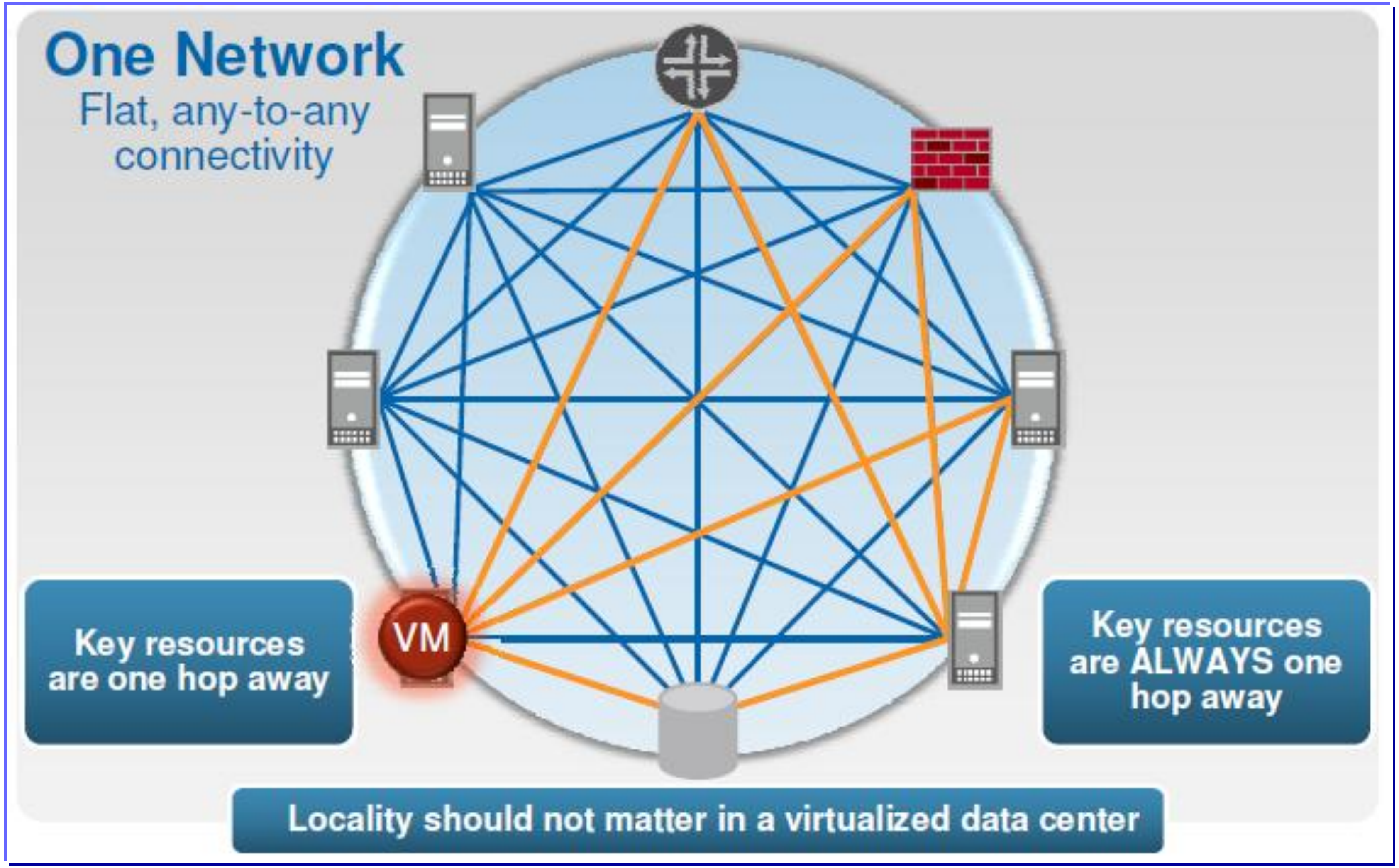
Major commercial vendors (e.g., Cisco, Juniper) are developing **Data Center Ethernet** or **One Network**, which uses layer 2 addresses for location independence and complex congestion control mechanisms

Target to reach

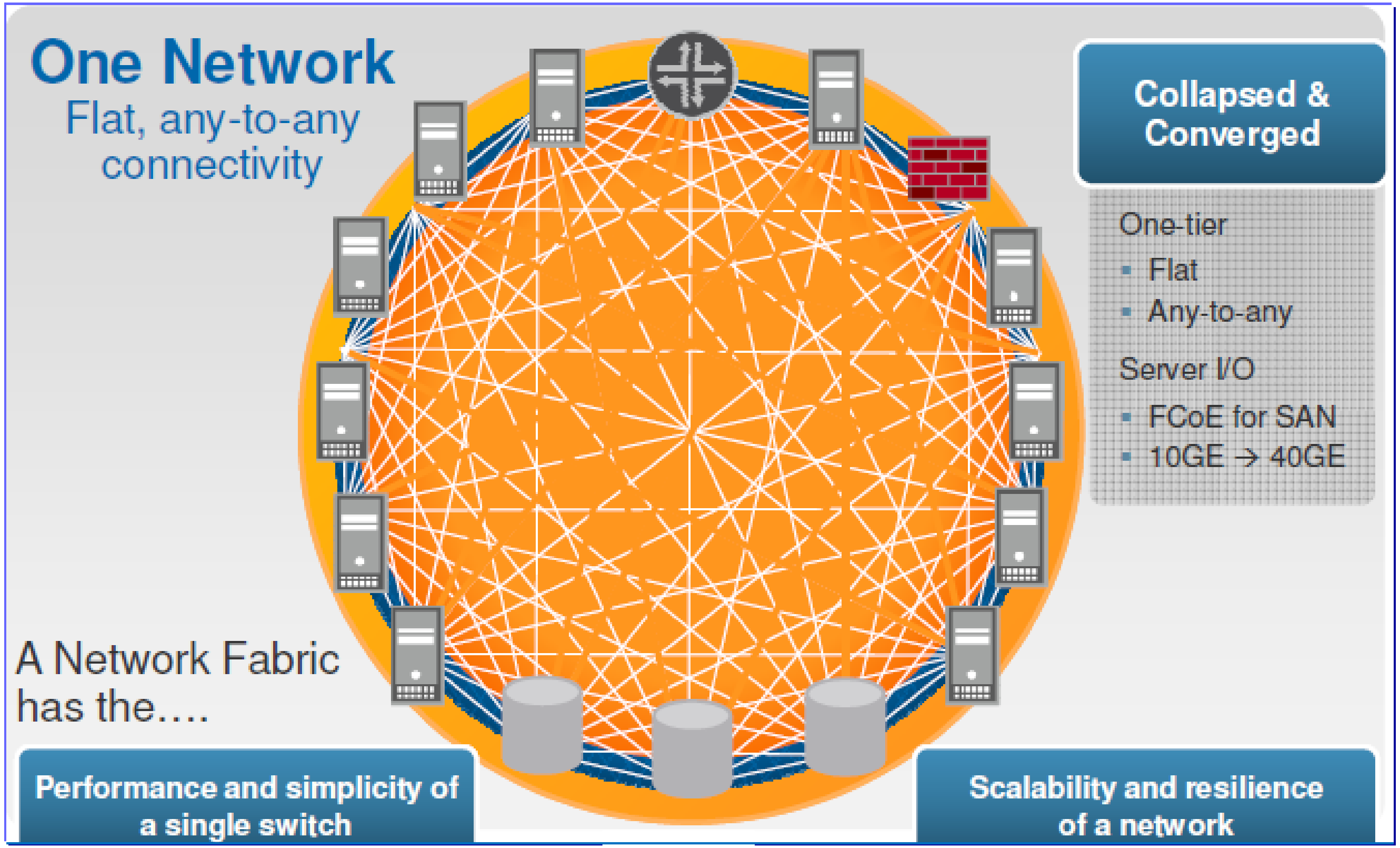




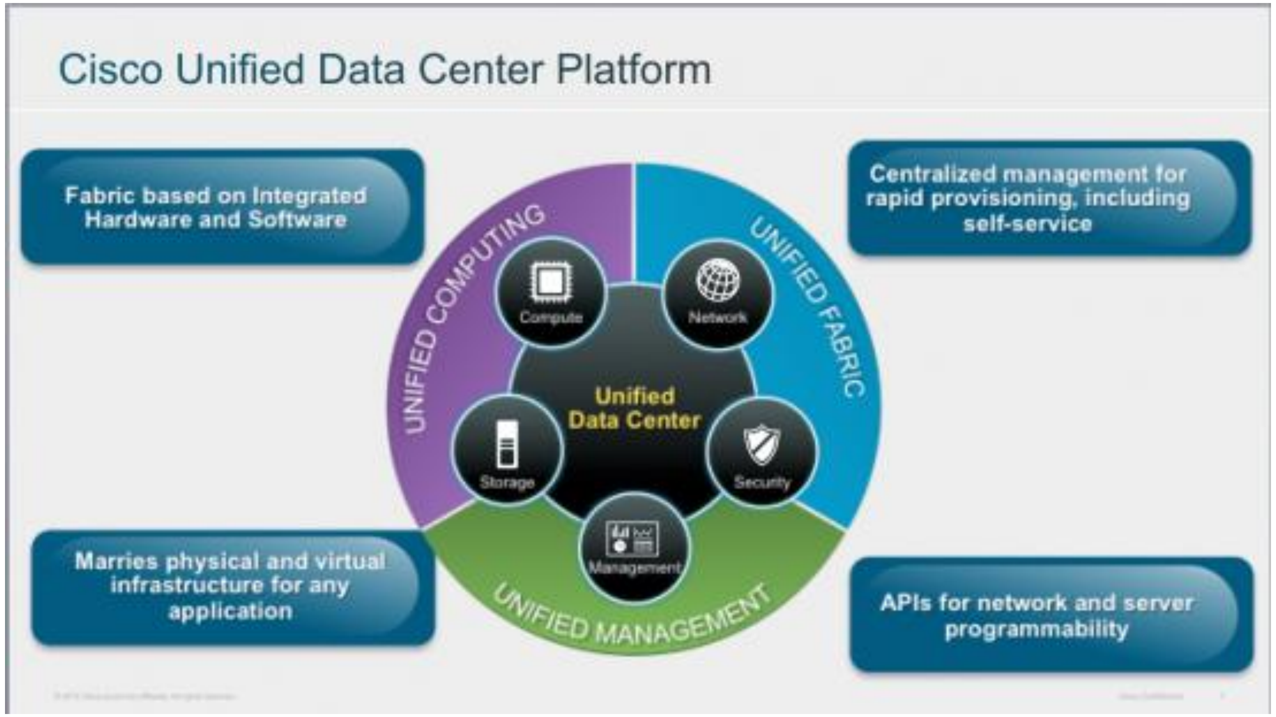
Locality should not matter

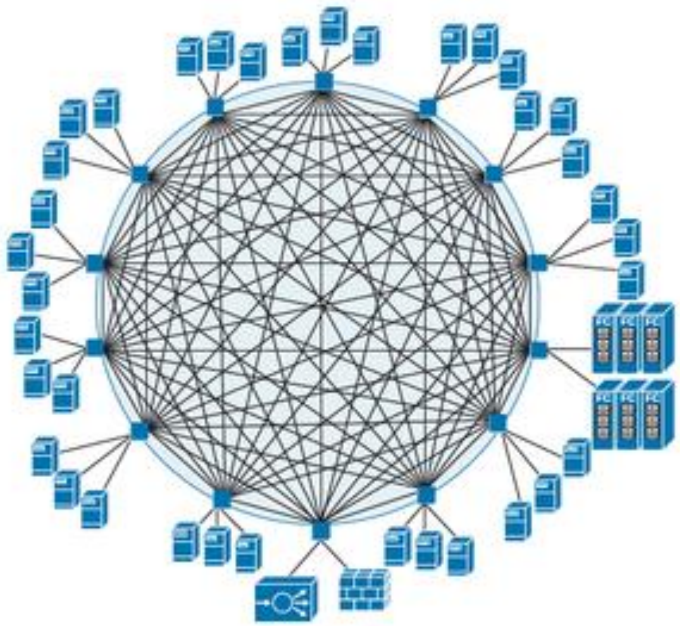


Datacenter Fabric (e.g., Juniper)



Unified Data Center Fabric by Cisco



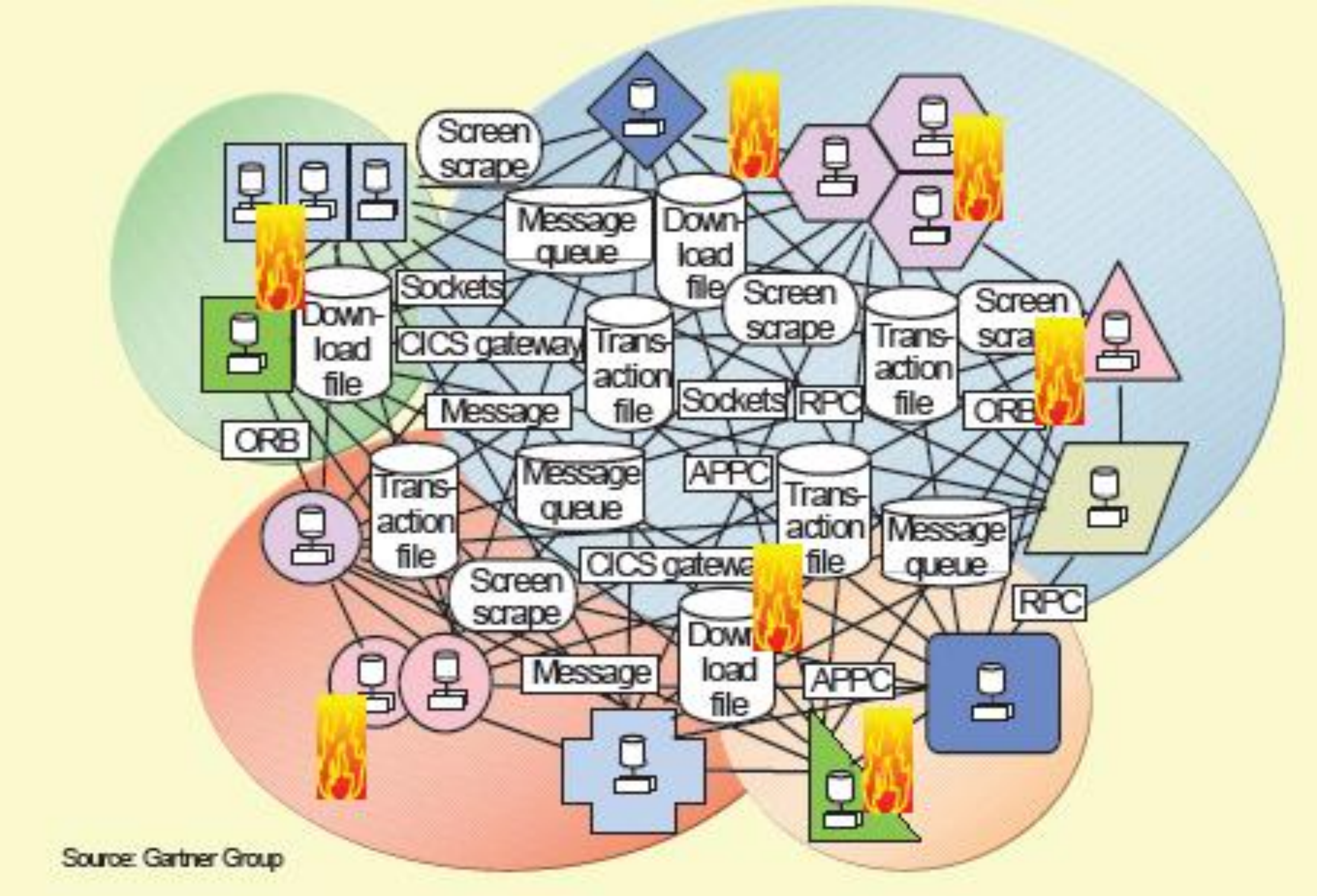


Hybrid architectures

- **Interconnect network**, **cloud**, and **multiple service providers**. → Aligning with a partner that can execute network and cloud interconnection seamlessly is valuable for IT leaders

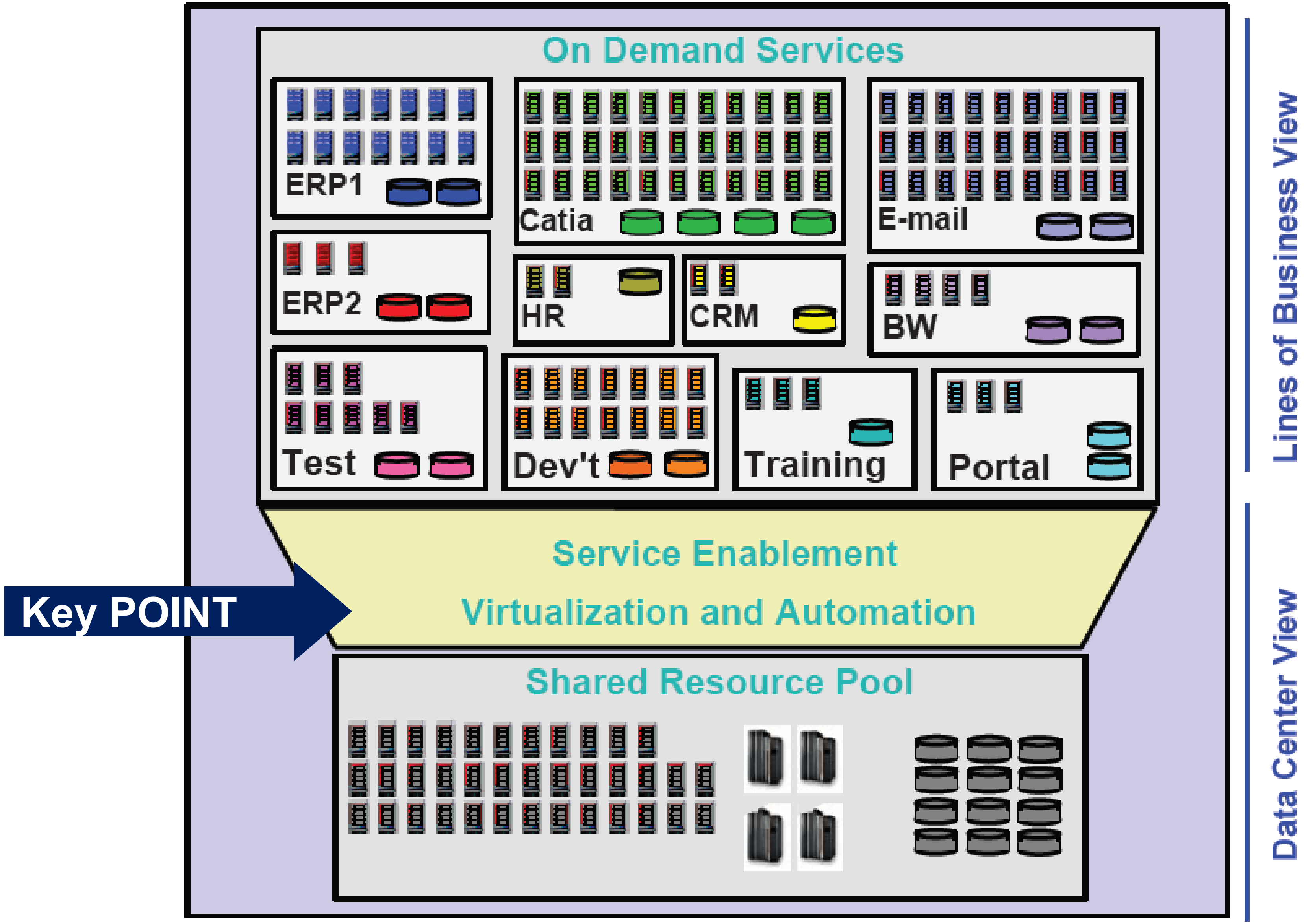
- Deploy, interconnect, and host **SDN edge**. In the high-stakes AI environment, having an airtight approach to deployment, network interconnectivity, and SDN is critical. Localization of these three components shores up seamless deployments with network reliability and security

Example of an “old” distributed infrastructure



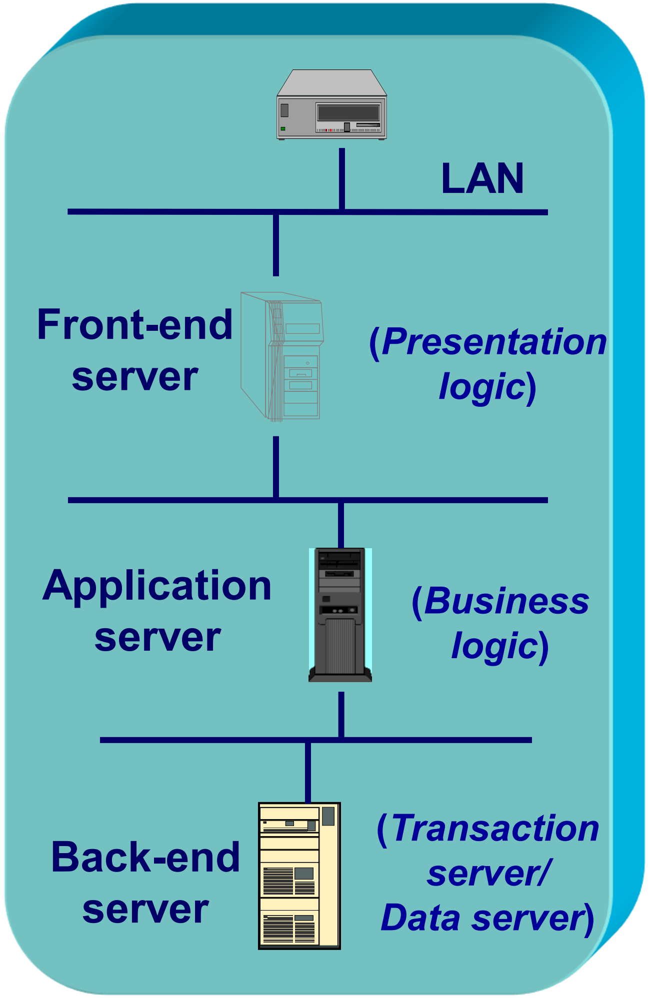
A similar architecture cannot scale, be reliable and not even secure. And yet there are still many of them!

The vision of a modern infrastructure (KISS)

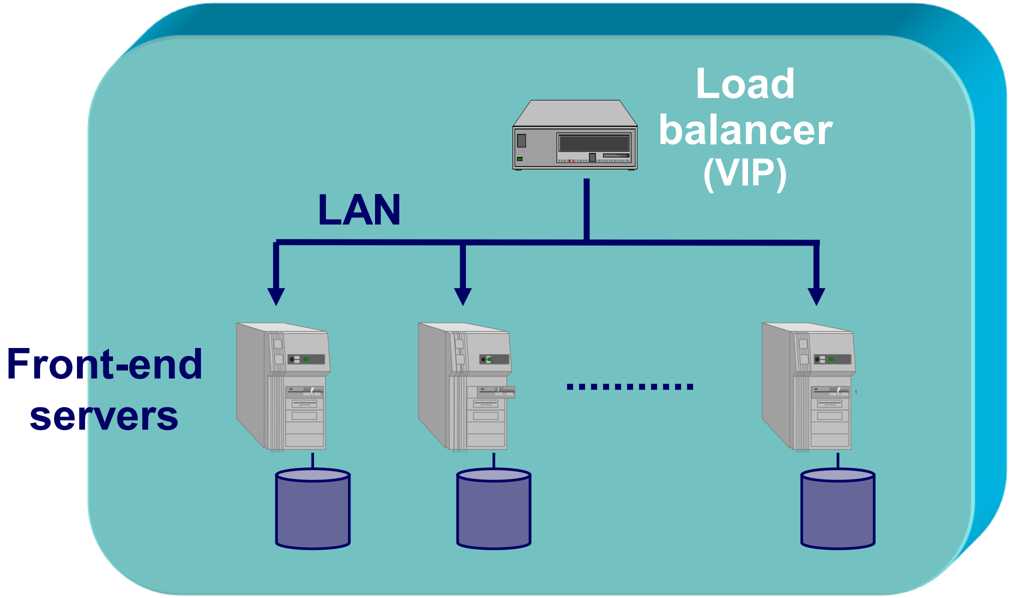


Server replication (*at local level*)

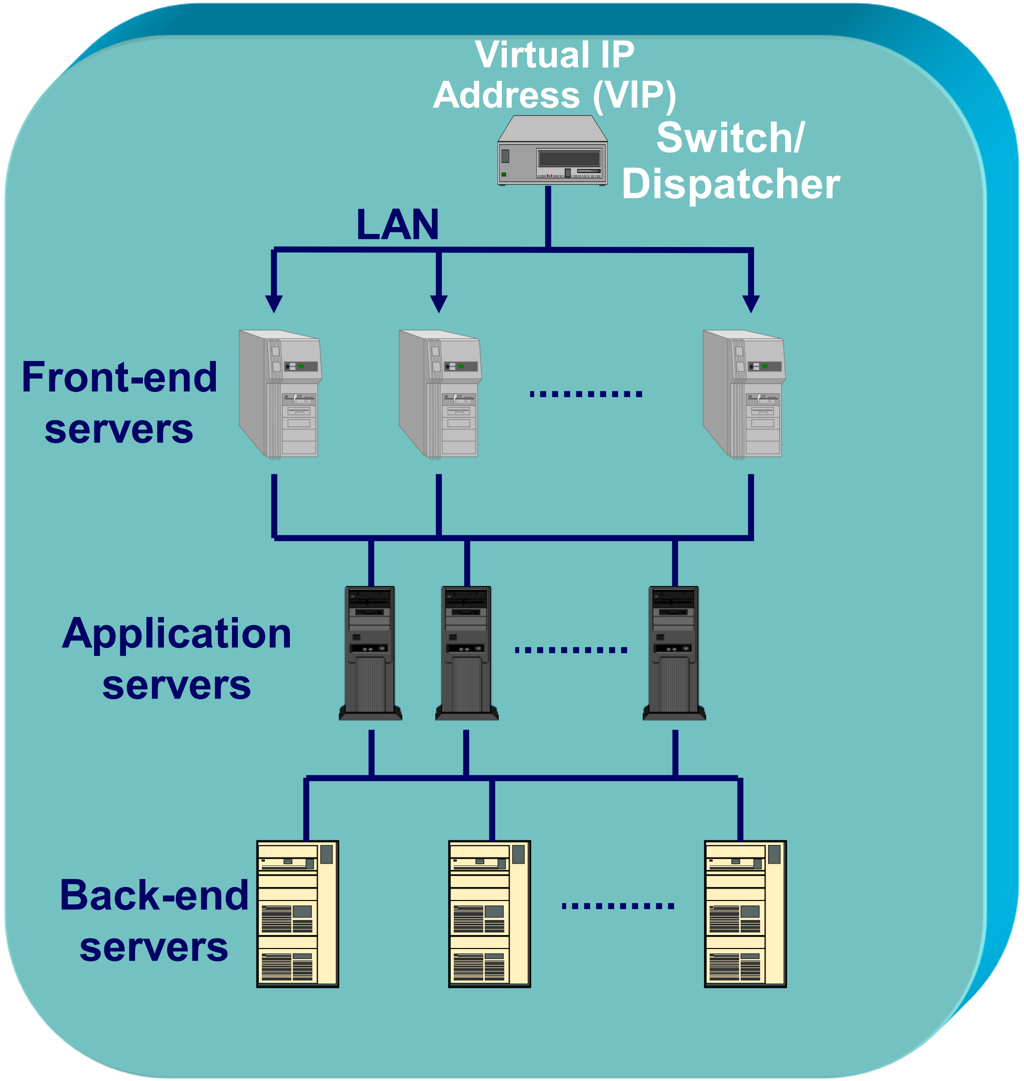
Vertical replication



Horizontal replication



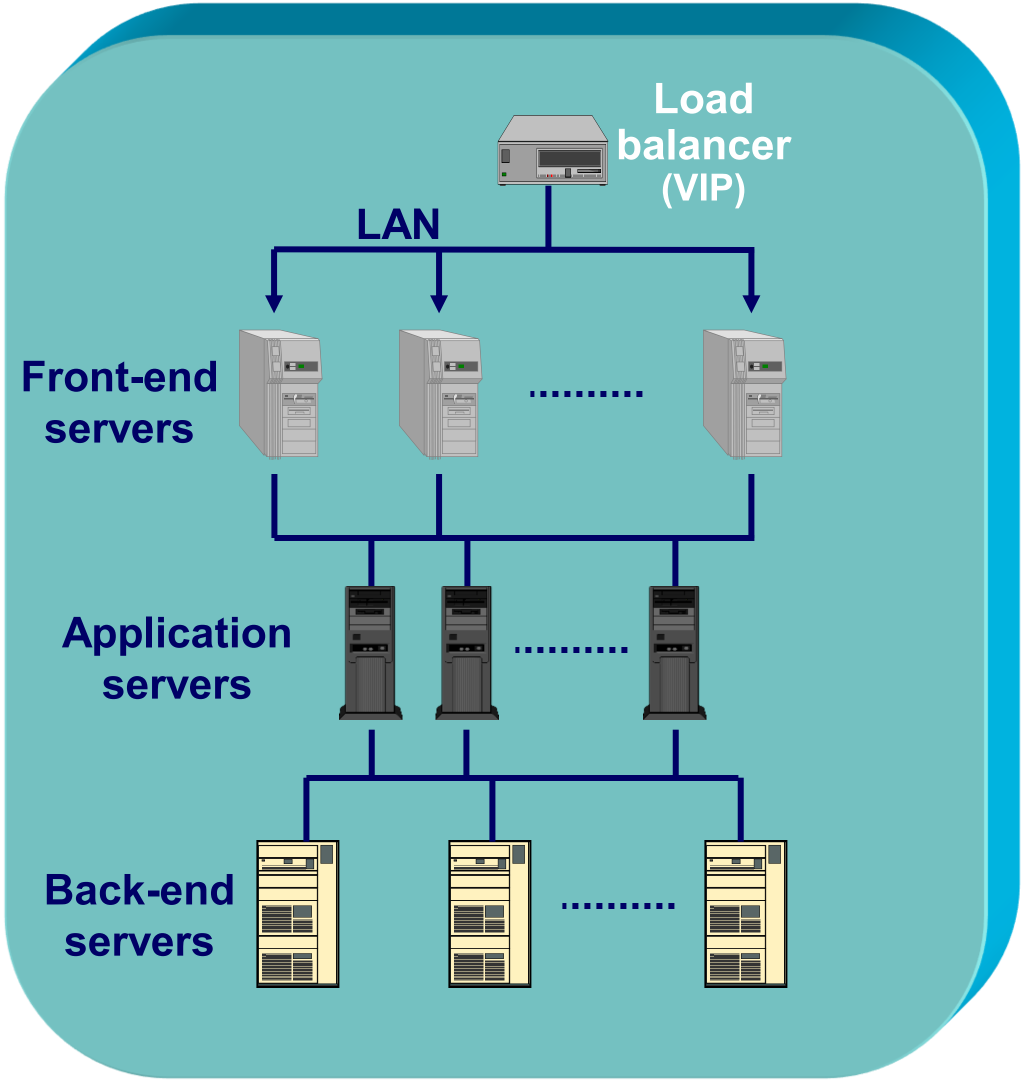
Combinations of Vertical+Horizontal replication



Most *Web-based services* are based on a logical aggregation of virtual machines in one or multiple Data center(s)

The first question for design

1) ***How many servers can we add?***



It depends on the logical level

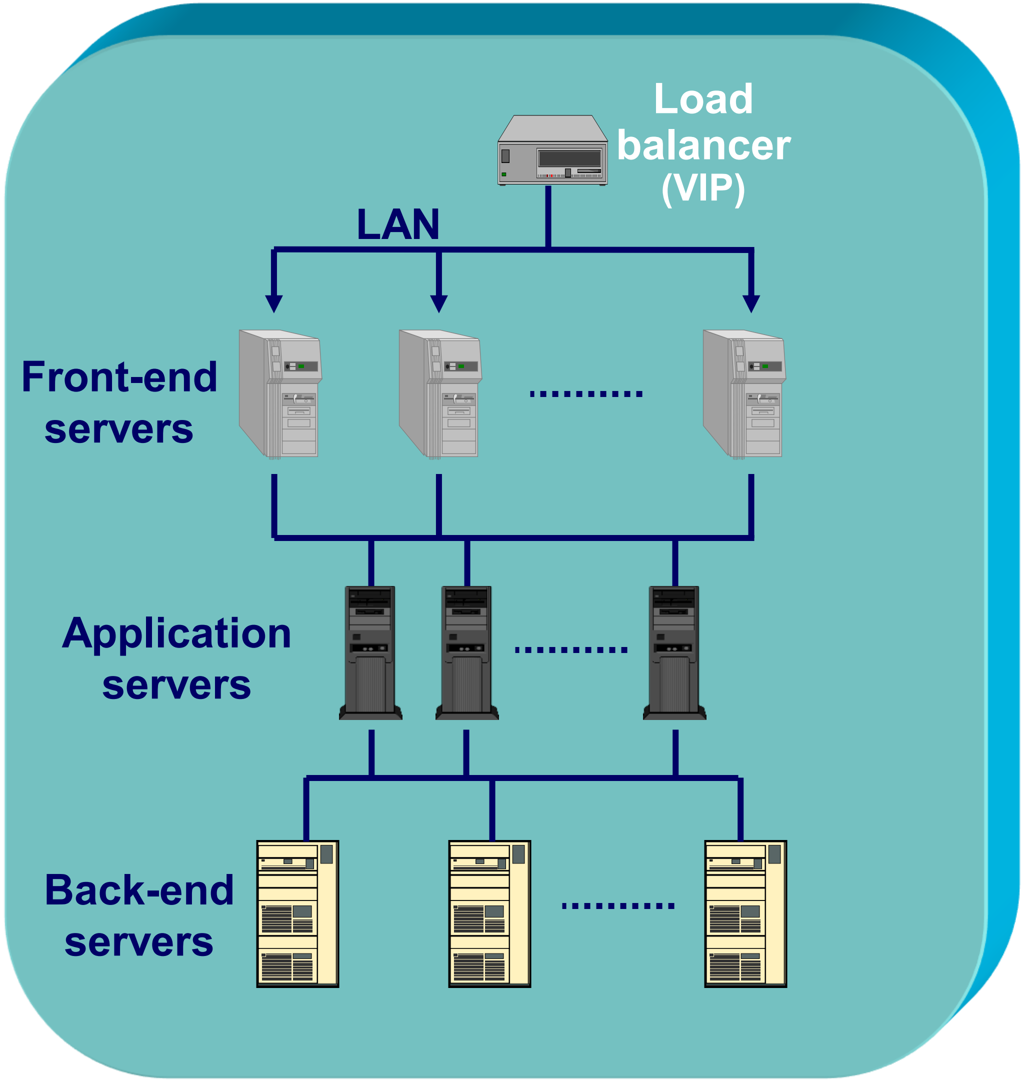
- **Presentation level**: No conceptual and no implementation limit because user requests are independent
  - Apache
  - Microsoft IIS
- **Application level**: Some limits can reside in the applications and mainly in their implementation
- **Data level**: Severe limits if there are many transactions (“write” operations). Few limits if there is a majority of “read” operations
  - Microsoft Sql server, IBM DB2, Oracle, MySql, Postgress

NO LIMIT ??
➔ INFINITE POWER ??

Second question for design

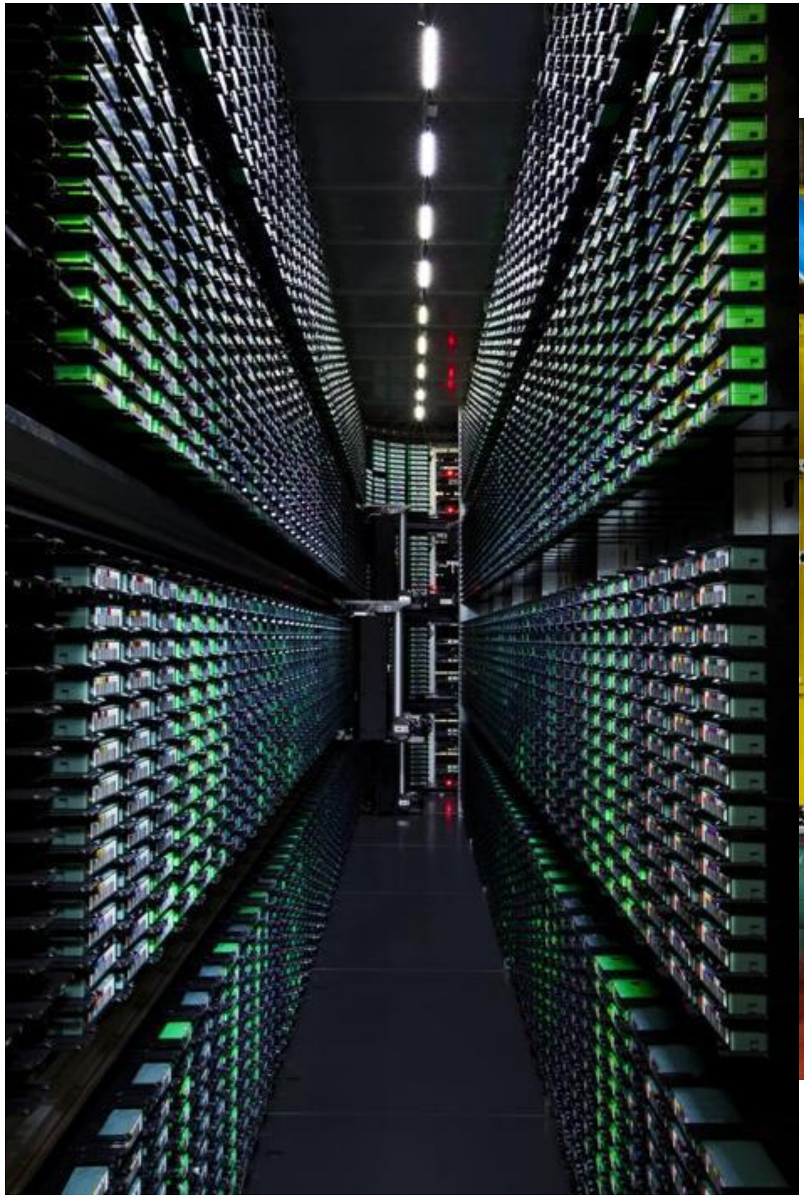
1) ***How many servers can we add?***

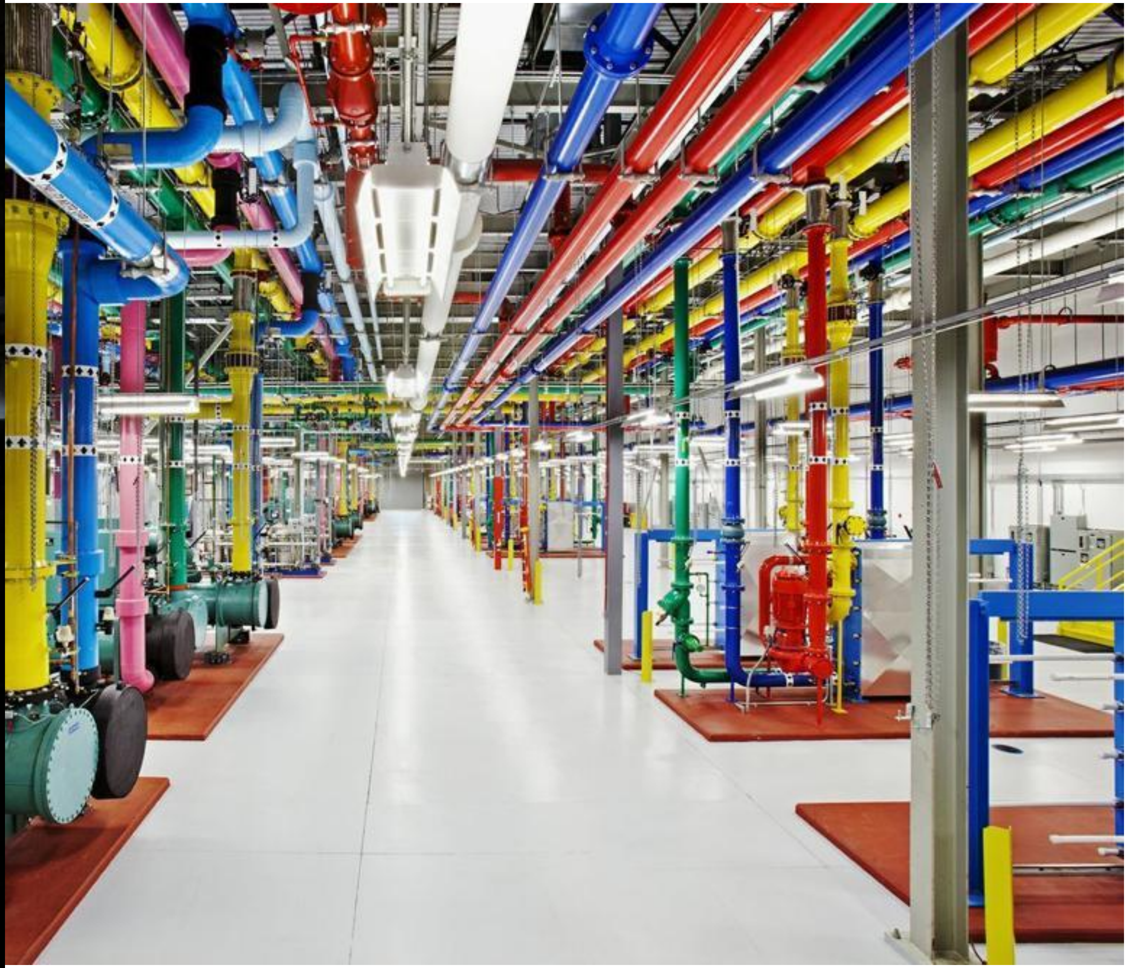
2) **How many servers is it convenient to add?**



Examples of real Internet Data Centers

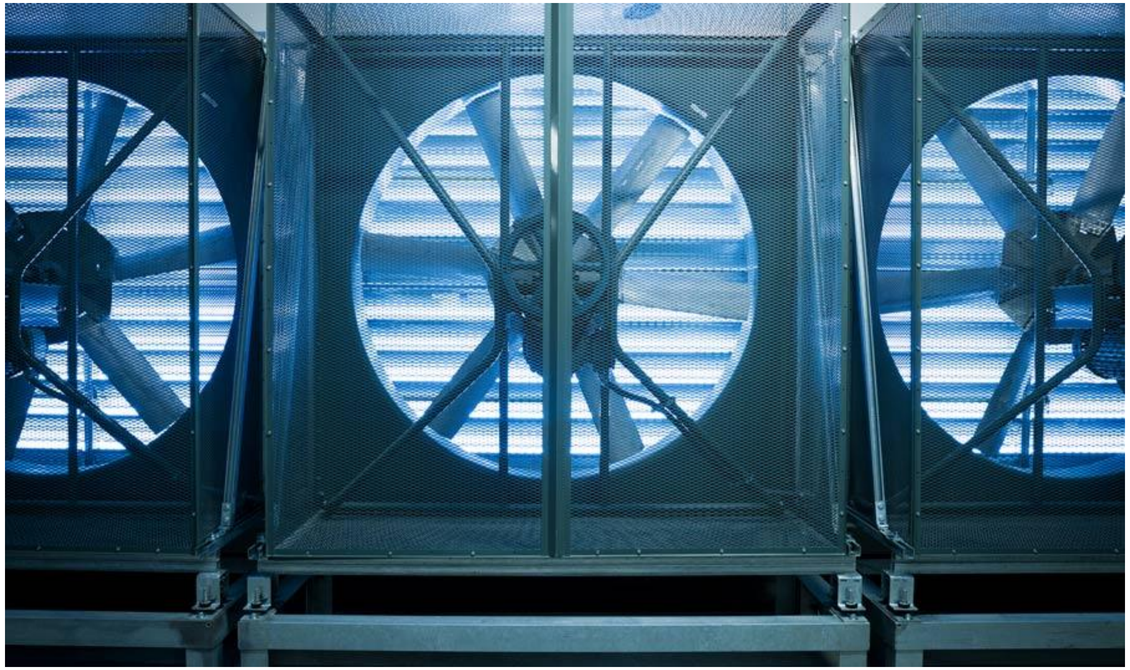
Google datacenter





Facebook datacenter

Facebook datacenter (*cooling system*)



Microsoft Azure

Apple (one iCloud center)

1 THE NEW QUESTION MARK
Is this the site of Apple's 4.8-megawatt biogas plant?

2 SubStation

3 Apple's 500,000 square foot data center. Home to the iCloud.

4 A smaller 21,000 square foot "tactical" data center.

5 The 100-acre solar farm will generate 20 megawatts of power.

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Apple

Italy (Milano): Aruba datacenter

Global Data Center

- Milan, Italy
- 100% green energy
- 90 MW
- Tier4
- 40.000m^2 in a 200.000m^2 campus

Italy (Bologna): ECMWF + Technopole

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An IDC should care not only of IT issues

- Interconnection to Internet (*first mile*)
- Architectures for logical security
- Logistics
- Cooling systems
- Anti-fire plant
- Physical security (access control, intrusion, alarm system)
- Detectors (smoke, water, temperature, humidity, …)
- Electrical systems and uninterruptible power supply (UPS)
- Disaster recovery plans
- **Site**
- **Size**

Strong commitment on security (1/2)

1. **PEOPLE**
   – Analysis of employees (past records and way of conduct, including strong logging of their actions)
   – Separation of duties
   – Security Team: 100+ experts
   – Incident Response Teams 24/7
2. **PHYSICAL**
   – Barriers, alarm, 24/7 monitoring, private policy checks, electronic and biometric accesses, CCTV, multiple generators, …
3. **LOGICAL**
   – Logical network security, host security, transmission level security, database security, AAL
   – Continuous monitoring of threats and vulnerabilities
   – Security embedded in the entire lifecycle software development

Strong commitment on security (2/2)

4. **ASSESSMENT and AUDITING**
   - Mitigate risks while complying with legal, statutory, contractual, and internally developed requirements

5. **POLICIES and PROCEDURES**
   - Develop and enforce policies and procedures at all levels
   - Design and secure information systems using security domains, defense in-depth and least privilege principles
   - Develop and integrate security architecture into business processes (CobiT, ISO27001) and achieve certifications: Safe Harbor, PCI, SAS70, FISMA, CSA, …
   - Conduct employee security awareness training classes
   - Perform regular vulnerability assessments and pen-testing (typically by internals)
   - External audits

Physical security

All entrances must be controlled

- badge or biometric systems
- prevent keys (except for emergencies)
- ports must be connected to an alarm system that must be activated if left open for over a period
- it is preferable to avoid windows

Disaster Recovery: to prevent and mitigate incidents

- smoke detectors and fire
- where possible, anti-flammable materials
- generators UPS at least 48 hours of power where operations must be tested periodically (even monthly)
- Redundant cooling system
- Backup system alternative far from the datacenter (>30Km is the rule)

Control room (h24 or 24/7)

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Data Runs the Planet –
Together we must ensure it does not ruin *the planet*

Sustainable data centers

- For years, companies that operate large data center infrastructures have been investing in “clean energy” sourcing as part of their strategies for becoming **carbon neutral** or **carbon negative**
- **Google Cloud** was the first major company to go *carbon neutral*. In 2018 it became the first major company to use 100% renewable energy

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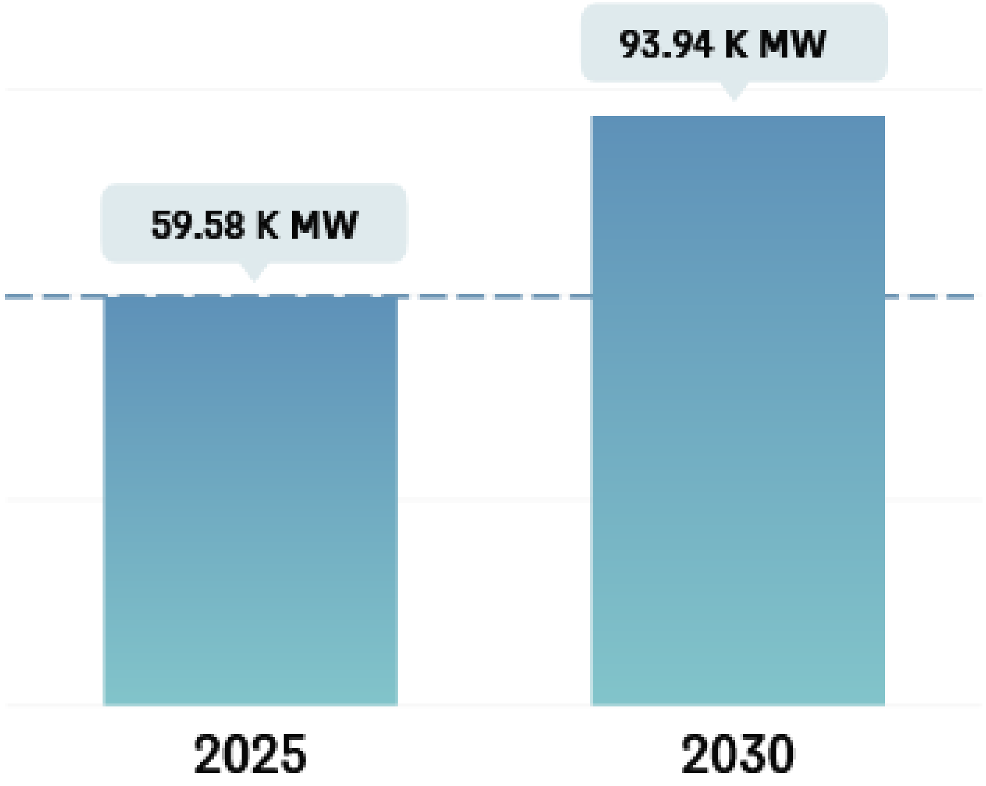
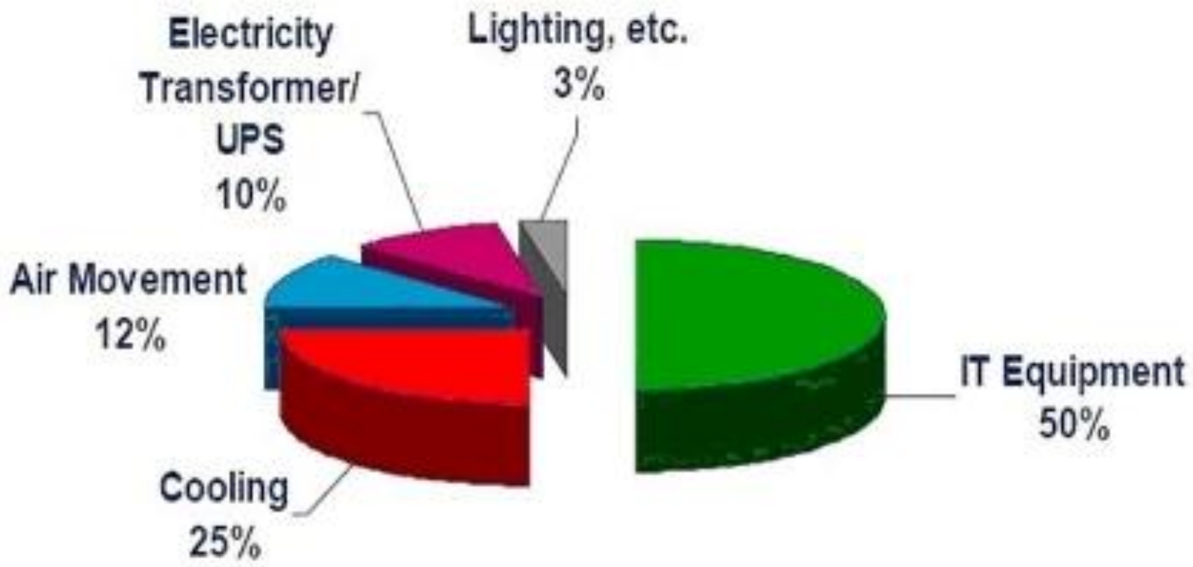
IDC: Growth of (total) energy consumption





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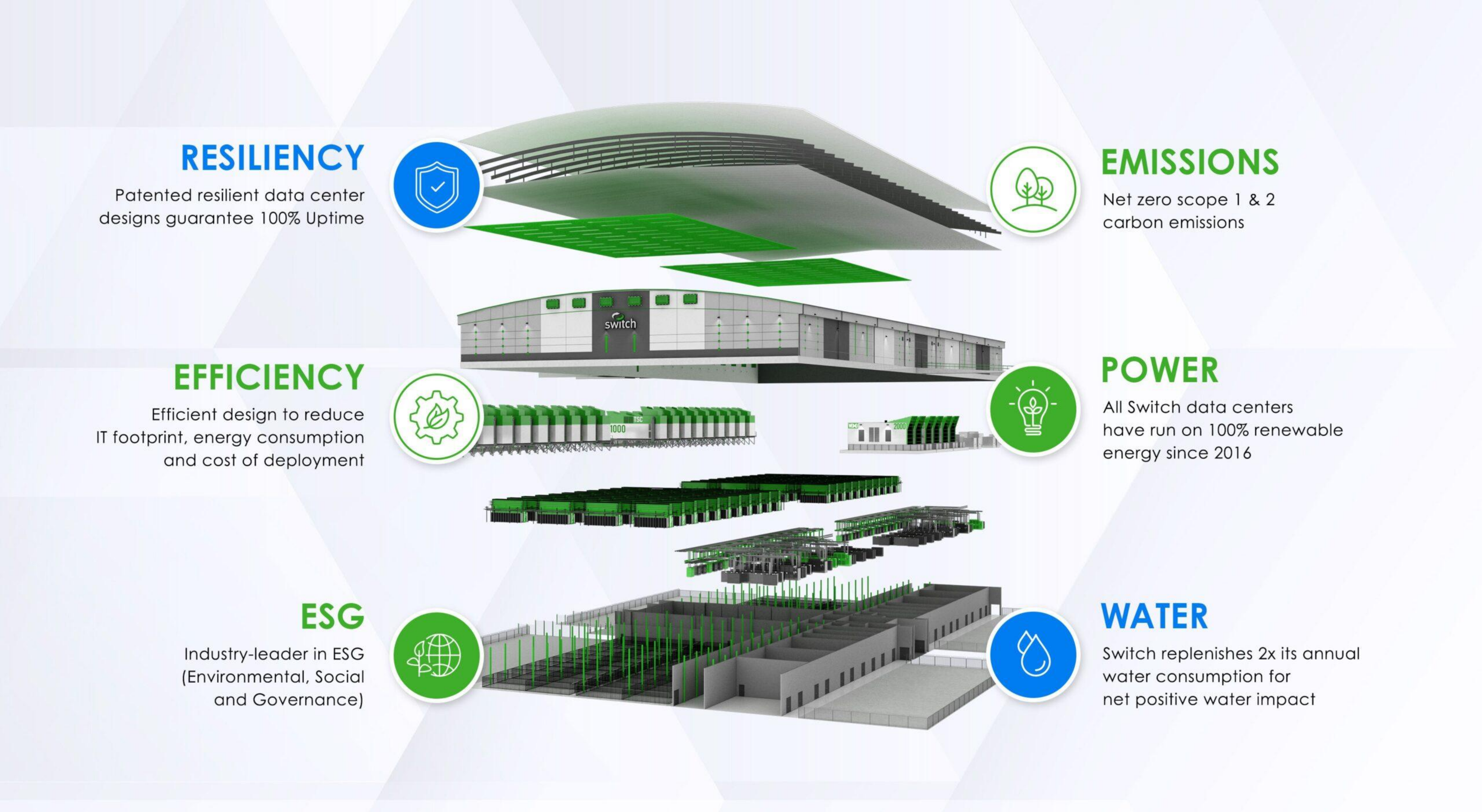
Sustainable data centers (1/2)



Sustainable data centers (2/2)

Renewable energy: Utilization of solar, wind, or hydroelectric power to reduce dependence on fossil fuels.

Efficient cooling: Implementation of techniques like liquid cooling, hot/cold aisle containment, and free air cooling to reduce energy usage.

Sustainable infrastructure: Use of eco-friendly building materials, energy-efficient lighting, and modular designs that allow for scaling without wasting resources.

Efficient hardware lifecycle management: Repurposing, recycling, or responsible disposal of IT equipment to minimize e-waste.

Virtualization: Maximizing server utilization to reduce the physical number of servers require

Sustainable energy and consumption

Increased investment by tech companies in next-generation techs that, if they fully pan out, could deliver much more significant energy and carbon savings than resources like solar and wind power

- **Intel** announced an effort to create a reference design for immersion cooling, a technique that involves immersing servers and other equipment in non-conductive liquids to enable hyper-efficient cooling
- There is a major interest among tech companies in leveraging hydrogen as a power source for data centers. The technology for making this happen on a large scale is not quite there yet
- **Amazon** committed to publishing *Water Usage Effectiveness* reports on a yearly basis. It also pledged to become water-positive, meaning it will make more water available than it consumes
- **Microsoft** and **Google** cloud datacenters have made similar pledges to water-positivity in recent years

Ranking

Data Center Players Strive For 100% Clean Energy

Green power share of top 10 data center operators

Data center company	Operational capacity	Capacity under development	Green power share
Amazon	3,414MW	10,849MW	100%
Microsoft	2,662	7,601	100
QTS	1,276	6,173	100
Meta	2,797	4,087	100
Google	2,823	3,681	100
Scala	114	5,127	100
Digital Realty	2,355	2,689	66
CloudHQ	341	3,575	Not disclosed
CyrusOne	934	2,211	62
Stack	560	2,564	Not disclosed

Source: DC Byte, BloombergNEF, company statements.
Note: Data as of end-2024. The green power share of Microsoft and QTS datacenters refers to 2025 targets. Data centers are ranked by the total IT capacity – both operational and in development (under construction, committed, early stage).

BloombergNEF

2nd Trend → Green data centers (*clean energy index*)

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Note: This list is subject to annual variation

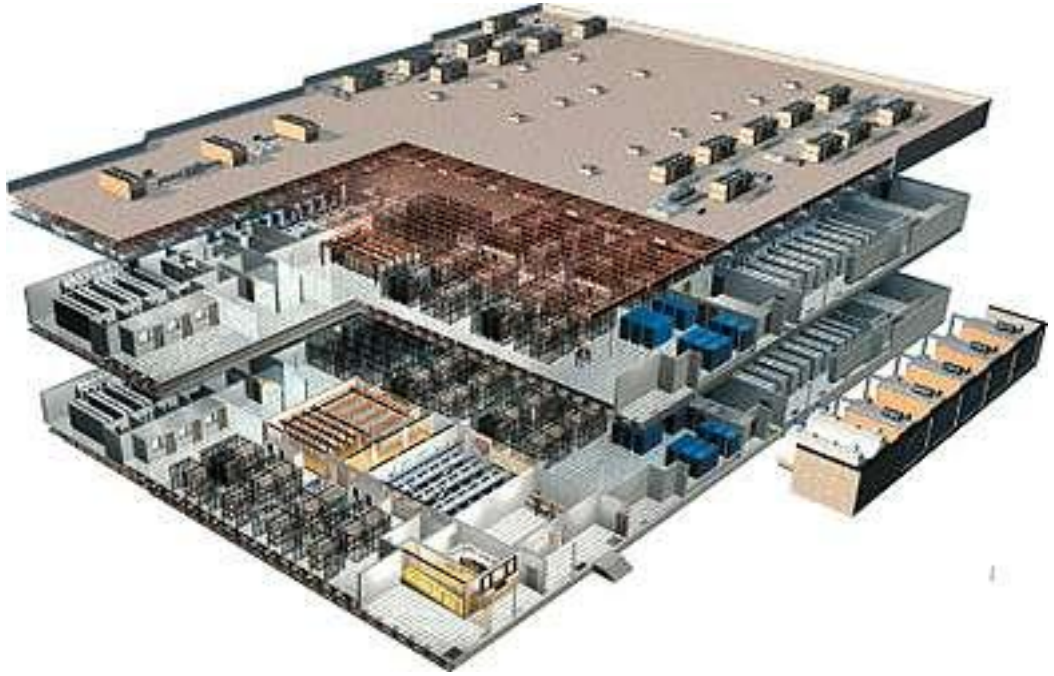
Company	Clean Energy Index	Coal	Nuclear	Energy Transparency	Infrastructure Siting	Energy Efficiency & GHG Mitigation	Renewables & Advocacy
Akamai	NA	NA		A	C	B	D
amazon.com web services	13.5%	33.9%	29.9%	F	F	D	F
Apple	15.3%	55.1%	27.8%	D	F	D	D
DELL	56.3%	20.1%	6.4%	C	C	C	D
facebook	36.4%	39.4%	13.2%	D	B	B	C
Google	39.4%	28.7%	15.3%	B	C	B	A
hp	19.4%	49.7%	14.1%	C	D	B	C
IBM	12.1%	49.5%	11.5%	C	D	C	D
Microsoft	13.9%	39.3%	26%	C	D	C	C
ORACLE	7.1%	48.7%	17.2%	D	D	C	D
rackspace HOSTING	23.6%	31.6%	22.3%	C	C	C	C
salesforce	4%	33.9%	31%	B	C	C	C
twitter	21.3%	35.6%	12.8%	F	D	F	D
YAHOO!	56.4%	20.3%	14.6%	C	B	B	B

Availability: Standard TIA-942

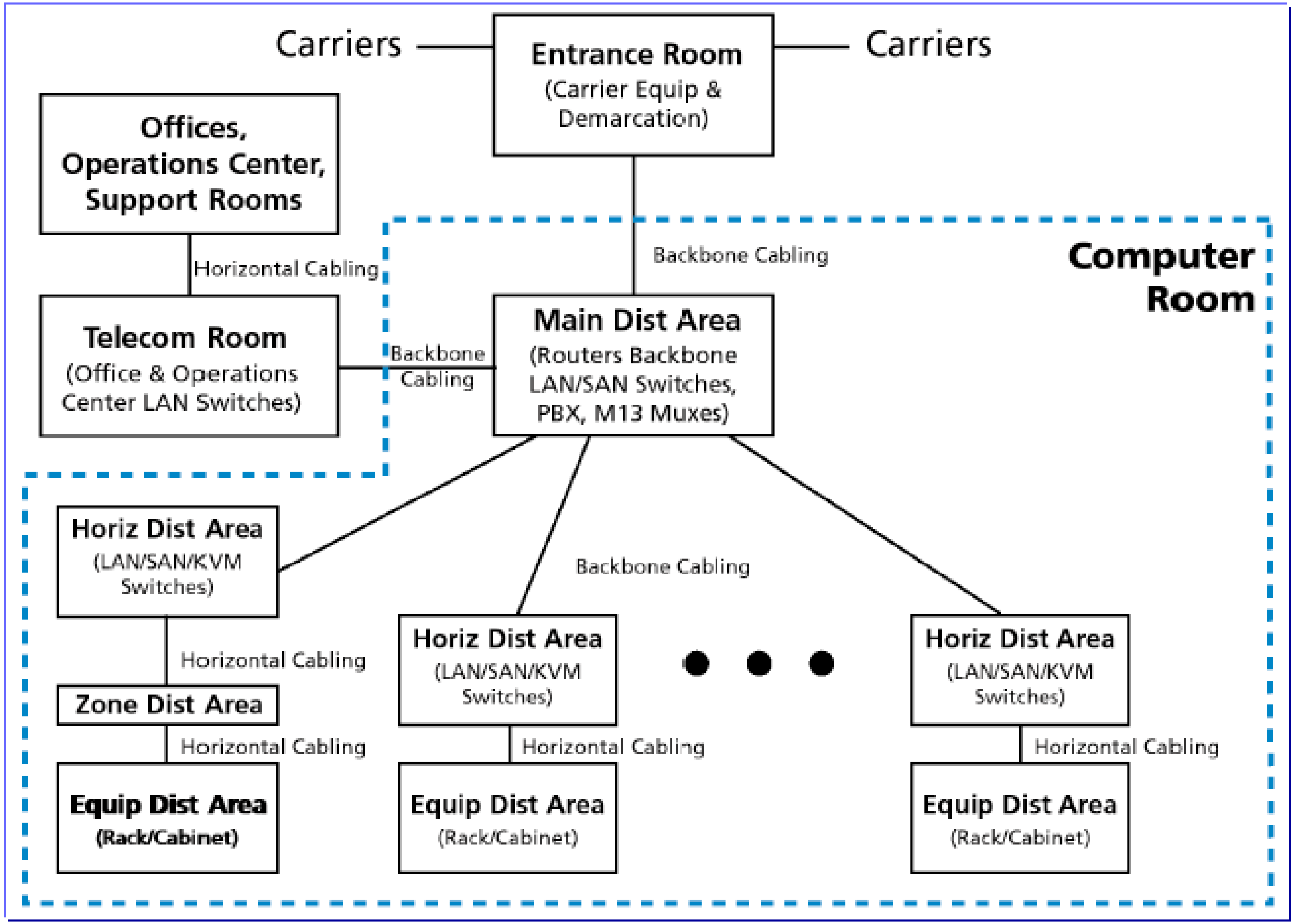
TIA = *Telecommunications Industry Association*

Several data center components

- Architecture
- Networking
- Power plant
- Cooling
- Mechanical parts
- Security
- Reliability
- …
- Costs



TIA complaint datacenter (key functional areas)



Tiered reliability for datacenters

Tier-1
- **99,671% availability** (max annual downtime: 28,8 hours)
- *Susceptible to service interruption from both planned and unplanned activity*
- Simple path for cooling and power distribution with no redundant component
- Implementation: 3 months

Tier-2
- **99,741% availability** (max annual downtime: 22 hours)
- *Simple path for cooling and power distribution with some redundant components*
- Implementation: from 3 to 6 months

Tier-3
- **99,982% availability** (max annual downtime: 1,6 hours)
- *Enables planned activities without interrupting services, but unplanned events may cause disruption*
- Multiple paths for cooling and power distribution with some redundant components
- Implementation: from 12 to 20 months – **Costs: 2 x Tier-1**

Tier-4
- **99,995% availability** (max annual downtime: 15 minutes)
- *Planned activities do not interrupt services, and data center can sustain at least one worst-case unplanned event with no critical impact on services*
- Multiple paths for cooling and power distribution with all redundant components
- Implementation: from 15 to 20 months

Glossary

Definitions (1/4)

- **Airside economizer** - A mechanical device used to take outside air into the data center when the air temperature is at or below the set cooling point
- **Colocation (retail)** - Placement of IT equipment in a provider's shared data center space. Targets smaller requirements for floor space and power requirements (typically less than 500 kW for total IT load). Colocation agreements can range from one to five-year terms and may include other items such as caged space or managed services
- **Container** - Originated from the use of an ISO (International Standards Organization) intermodal shipping container being used to accommodate data center racks. Many variations of the 20 and 40 foot long containers exist and fulfill needs such as quick deployment and mobility. Power and cooling plants are also available in ISO shipping containers
- **(hot or cold aisle) Containment** - Placing isolation barriers at the front, back and top of aisles to eliminate air mixing
- **DCIM (Data Center Infrastructure Management)**. The integration of IT and facility management disciplines to centralize monitoring, management and intelligent capacity planning of a data center's critical systems

Definitions (2/4)

- **Disaster Recovery** - The process, policies and procedures related to preparing for recovery or continuation of technology infrastructure critical to an organization after a natural or human-induced disaster. Disaster recovery is a component of a Business Continuity Plan (BCP) which focuses on aspects of a business functioning in the midst of disruptive events
- **Evaporative cooling** - Also known as swamp cooling, this is a way to cool air that takes advantage of the drop in temperature that occurs when water that is exposed to moving air begins to vaporize and change to gas
- **HPC - High Performance Computing**. The use of parallel processing for running advanced application programs efficiently, reliably and quickly. The term applies especially to systems that function above a teraflop (10^{12} floating-point operations per second)
- **IRR** - The Internal Rate of Return on an investment project is the annualized effective compounded return rate' or discount rate that makes the net present value (NPV) of all cash flows (both positive and negative) from a particular investment equal to zero
- **ISP** - Internet Service Provider

Definitions (3/4)

- **ISO/IEC 20000** - The first international standard for an IT Service Management System. It is based on and supersedes BS 15000
- **MSP** - Managed Service Provider
- **MicroGrid** - A localized grouping of electricity generation, energy storage, and loads that normally operates connected to a traditional centralized grid (utility) connected to it - generators, consumers and those that do both – in order to efficiently deliver sustainable, economic and secure electricity supplies
- **Modular** - A conceptual definition for a method of deploying components of, or a complete data center. Modularity implies a pre-packaged, sometimes manufactured module of data center components that through its standardized and pre-defined parameters enable scalability and rapid delivery schedule
- **PUE (Power Usage Effectiveness)** - A metric developed by The Green Grid (thegreengrid.org) to determine the energy efficiency of a data center. PUE is calculated by dividing the amount of power entering a data center by the power used to run the IT load within it
- **REIT (Real Estate Investment Trust)** - A security that sells like a stock on the major exchanges and invests in real estate directly, either through properties or mortgages

Definitions (4/4)

- **UPS (Uninterruptible Power Supply)** - The UPS is intended to provide clean undisturbed stabilized AC voltage, within strict amplitude and frequency limits, to protect the IT load from any utility power disturbances and irregularities, including outages for a limited time dictated by the capacity of the battery bank. The term UPS refers generally to AC Static systems, other types include DC and Rotary UPS

- **Water Economizer** - A mechanical device used to take cold air and cool an exterior water tower. The water from the chilled tower is then used in the air conditioners inside the data center instead of mechanically chilled water

- **Wholesale data center service** - Leasing of a fully-built data center space. The wholesale tenant gains more control over how their space is used for security and supporting infrastructure. Having the built-out space that is supplied with electrical and environmental infrastructure allows the tenant quick time-to-market in getting their IT infrastructure up and running

Placement of an IDC

Where to place your IDC?

- XCZXC

Factors influencing where to place data centers

Placement and sizing of an IDC is an interesting and challenging optimization problem including ***technical*** and ***management*** factors:

1. **Zoning**
   - cheap land and space for future expansions
   - safe land
   - proximity to stores operating 24 hours a day
2. **Networking**
   - presence of large POPs
   - excellent (multiple) backbones
3. **Energy**
   - proximity to abundant and cheap electrical services
   - power concessions
4. **County and city policies**
   - incentives
   - tax break policies
5. **Availability of human competences**

Locations

County	State	Data Centers Operating and In Construction (MW)	Planned (MW)
Loudoun	VA	5929.67	6349.4
Maricopa	AZ	3436.1	5966
Prince William	VA	2745.4	5159
Cook	IL	1478.13	2001.8
Milam	TX	1442	0
Santa Clara	CA	1314.73	552.5
Dallas	TX	1294.58	2911.2
Franklin	OH	1257.4	483
Morrow	OR	1177	0
Umatilla	OR	1118.5	101
Mecklenburg	VA	1019.5	502.5

Why are Data Centers located where they are?

It is tempting to assume that large populations drive data center development, but that is often not the case. Instead, it comes down to a mix of factors:

1) **Electricity availability and cost:** Data centers consume vast amounts of power. States like Virginia, Texas, and Oregon offer competitive electricity pricing and stable infrastructure

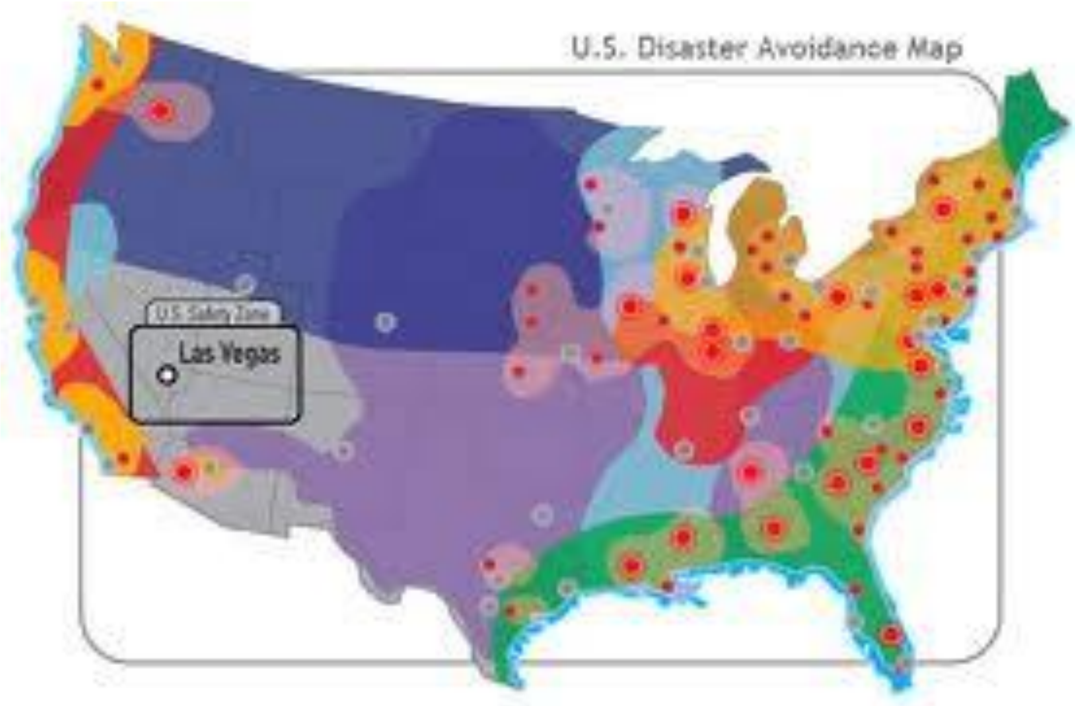
2) **Access to water:** Many facilities use water for evaporative cooling, so proximity to aquifers or rivers can be crucial

3) **Fiber networks:** Low latency is king. Proximity to fiber optic infrastructure and subsea cable landing stations is critical

4) **Zoning and incentives:** Tax incentives and permissive local zoning laws can make or break a deal

Zoning: safe land

The datacenters should be located in the #1 rated geographically safety zones

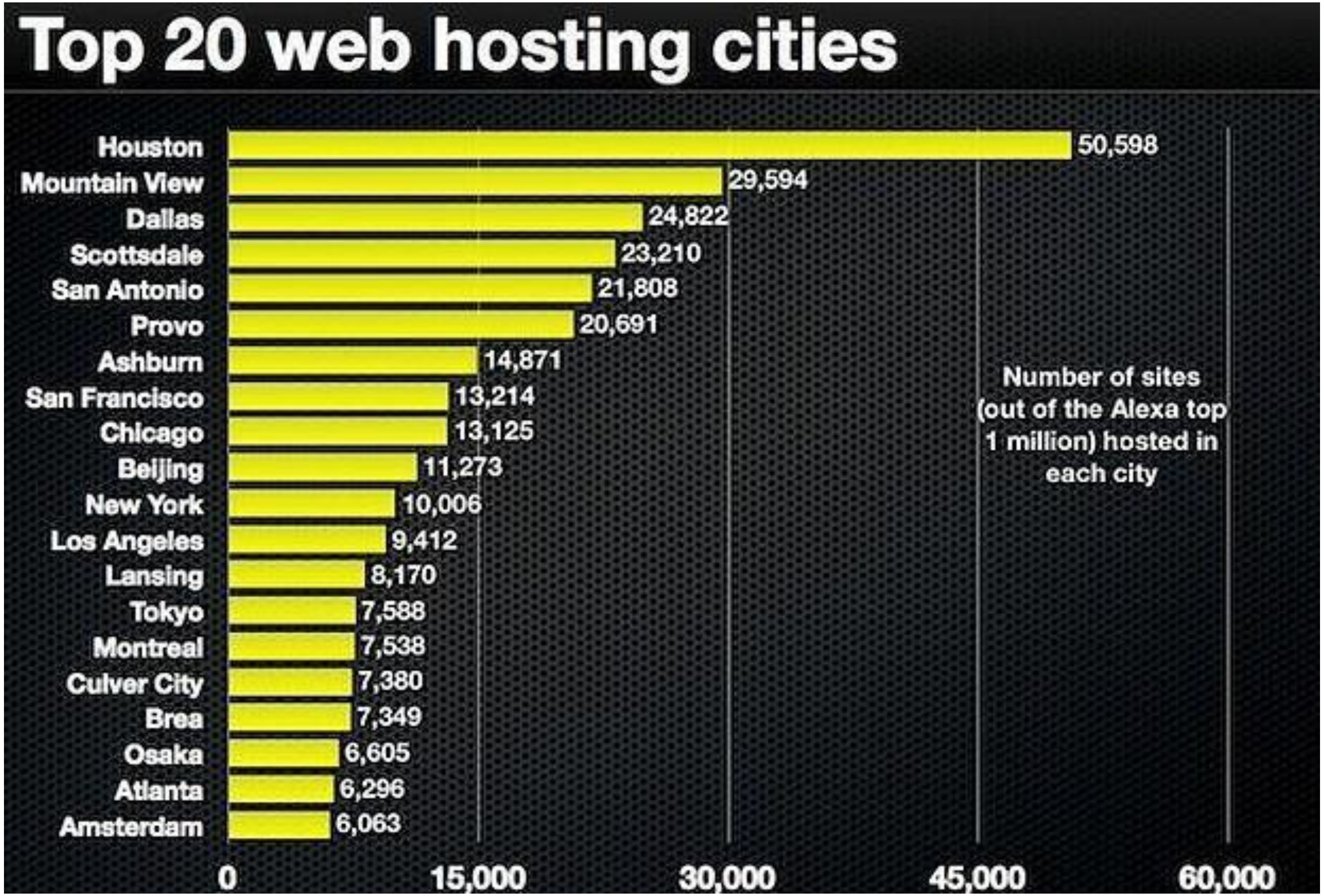


Networking: IDCs tend to flock together

- **The interesting but unsurprising fact about datacenters is that they tend to be in the same areas**

- There will usually be some anonymous buildings containing millions of dollars of hardware
- So, when the big players of the world will decide to hook up, they will find that it is relatively cheap: “often their counterparts are just across town”

Top 20 hosting cities



Ex., SuperNAP near Las Vegas

Energy sources: *examples*

Google in Oregon

Google in Dallas

Microsoft is building “Prometheus” supercluster is a **1 Gigawatt** site

Oracle and **NVIDIA** are building “Stargate I” in Abilene, Texas, which is designed to house over 450,000 Nvidia GPUs

China Telecom: Inner Mongolia Information Park

Google datacenters
https://www.youtube.com/watch?v=9CL3pZfsHbs

Colossus 1 and 2
https://www.youtube.com/watch?v=RxuSvyOwVCI

Musk’s xAI

xAI is investing more than **$20 billion** in Mississippi to build a massive data center, called **MACROHARDRR**

This 800,000 sqft facility will host the world’s largest supercomputer and mark the largest investment in state history

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Site - AVOID

- Risk of seepage water from outside or from broken pipes indoor → Ancient and wrong choice: to locate the datacenter at ground level or even below)
- Vapor and moisture infiltration
- Direct irradiation of sunlight
- Not monitored entrances through other areas
- Adjacency to tanks and diesel generators
- Lack of facilities for logistics, truck access, lifts, elevators, etc.

Typical IDC issues in one year

- ~0.5 overheating (power down most machines in <5 mins, ~1-2 days to recover)
- ~1 PDU failure (~500-1000 machines suddenly disappear, ~6 hrs to come back)
- ~1 rack-move (plenty of warning, ~500-1000 machines powered down, ~6 hrs)
- ~1 network rewiring (rolling ~5% of machines down over 2-day span)
- ~20 rack failures (40-80 machines instantly disappear, 1-6 hours to get back)
- ~5 racks go wonky (40-80 machines see 50% packet loss)
- ~8 network maintenances (4 might cause ~30-minute random connectivity losses)
- ~12 router reloads (takes out DNS and external vIPs for a couple minutes)
- ~3 router failures (have to immediately pull traffic for an hour)
- ~dozens of minor 30-second blips for DNS
- ~1000 individual machine failures (2-4% failure rate, machines crash at least twice)
- ~thousands of hard drive failures (1-5% of all disks will die)

In addition, software bugs, configuration errors, human errors, …

Tips for performance and resiliency

1. **Check for bottleneck**

2. **Check for single points of failure**

3. **Keep it simple and then scale** (**KISS**): the reason many systems use centralized control

4. **If it is not well tested, do no rely on it**

Question: how do you test availability with hundreds of loosely coupled services running on thousands of machines?

In summary

Technique	Performance	Availability
Replication	✔	✔
Partitioning (sharding)	✔	✔
Load-balancing	✔	
Watchdog timers		✔
Integrity checks		✔
Canaries		✔
Eventual consistency	✔	✔

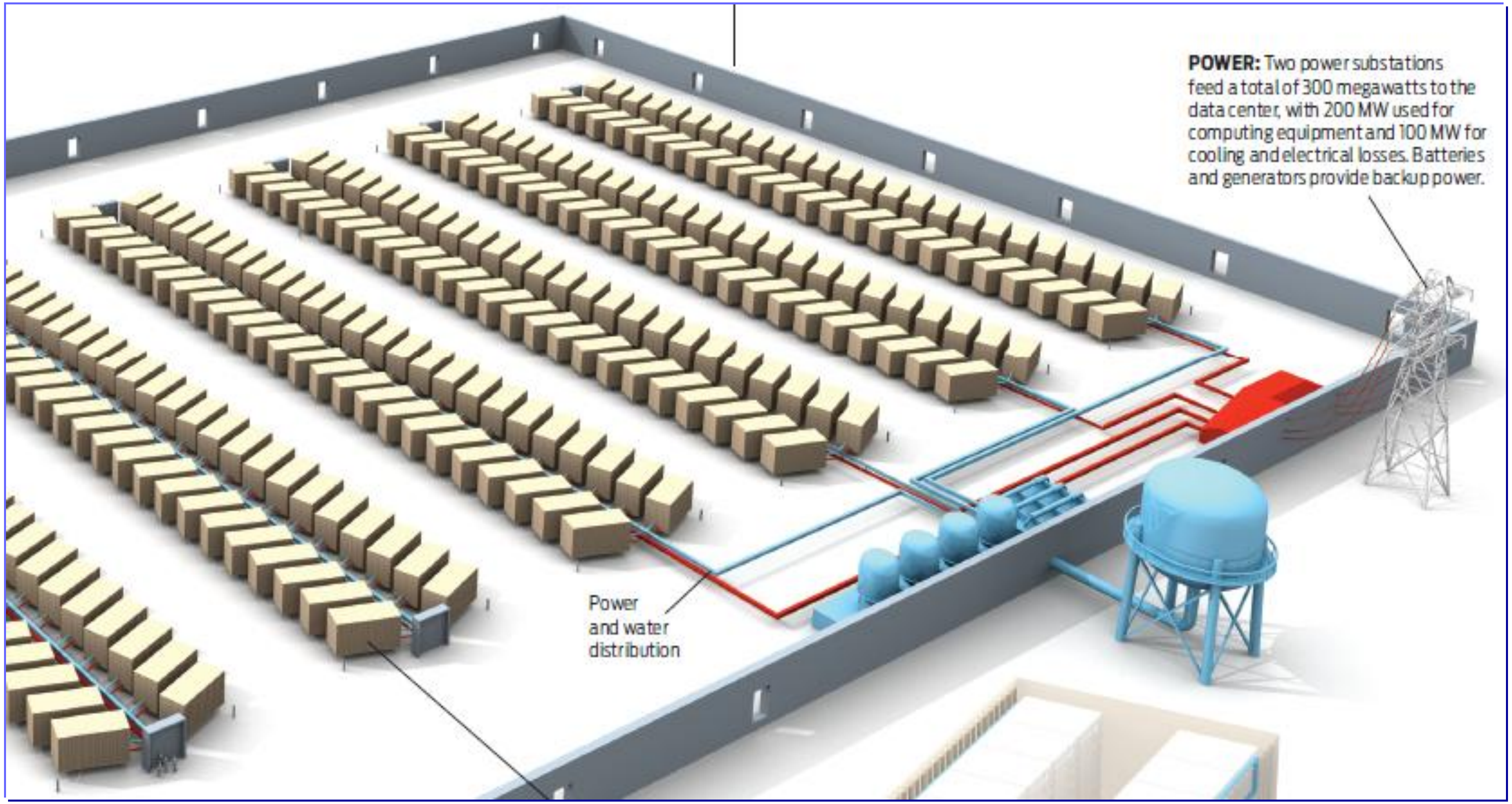
1st Trend → Mega data-centers

- Large providers need some number of mega-datacenters to house large computations
- The size of a mega-datacenter is typically determined by **extracting the maximum benefit from the economies of scale** available at the time the data center is designed
- This is an exercise in jointly optimizing server cost and power availability:

Typical design →

100,000s of servers spread over 100,000s of square feet drawing from 10 to 20MW of power

“The million server datacenter”



Why incentives and competition among cities?

AWS Has Spent $35 Billion on its Northern Virginia Data Centers

Amazon Web Services (AWS) has spent $35 billion on its cloud computing infrastructure in Northern Virginia over the past 10 years, illustrating the enormous impact data centers can have on regional economies.

Rich Miller

Oct. 4, 2021

Google to spend $9.5B this year on data centers and offices

Investing in offices is "more important than ever," Google CEO Sundar Pichai said, even as the company embraces hybrid work

Musk's xAI raises $20 billion in upsized Series E funding round

By **Reuters**

January 6, 2026 9:53 PM GMT+1 · Updated January 6, 2026

Costs (1/2)

- A data center of the size that Facebook or Google might use would cost $500 million
- The two new Amazon data centers north of Jackson, announced a year ago, were originally estimated to cost $10 billion but costs have now increased to **$16 billion**. Amazon called it “the single largest capital investment in Mississippi’s history”
  - Details: Amazon plans to spend $7.2 billion on servers, fiber-optic cable, and tech infrastructure for the Mississippi project, per state planning documents.
  - **The company estimated the investment would add some $3.3 billion to the state’s gross domestic product** and create 1,000 jobs by 2027

Costs (2/2)

- **Cost overload:** Data center expansion costs are escalating due to the need for expensive hardware like **Nvidia's** GPUs and other IT infrastructure. That could also affect estimates for the $500 billion **Project Stargate**
- Gartner: **Data center spending is expected to rise 15.5% YoY**, but we are now seeing that estimate is conservative
- **Key takeaway**
  - Costlier data centers could raise prices for web hosting, video and audio streaming, ecommerce, AI, and apps and services
  - **These costs will likely be passed down to businesses and consumers** relying on these services and could result in them seeking cheaper alternatives or canceling subscriptions and services. Budgeting for those changes in advance will likely be the **smart play**